Mathematics Exercises

2026

ESQC mathematics exercises

2026 edition

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Preface

This is a collection of mathematics exercises for the European Summerschool in Quantum Chemistry (ESQC).

Topics needed (List from Lucas)

If an exercise is provided, please cross it out below.

- Second quantization
  - ☐ Binomial expansion
  - ☒ binomial coefficients
  - ☒ Pascal's triangle
  - ☐ Cauchy product of series
  - ☐ Proof by induction (or direct)
  - ☒ Handling summation signs (also in combination with Kronecker deltas)
  - ☐ Definition of exp as a power series
  - ☐ Partial derivatives of vectors (CI eigenvalue problem)
  - ☐ Finding eigenvalues and eigenvectors of a 2x2 matrix.
- SCF
  - ☒ Matrix algebra: matrix multiplication
  - ☐ unitary transformation
- DFT
  - ☐ functional derivatives
  - ☐ Transformation formula for multidimensional integrals (calculating Jacobi determinant for transformation from one set of coordinates to another).
- Response theory
  - ☐ BCH expansion, manipulation of exponentials of operators.

- ☐ Exponential of antihermitian operator is unitary (and maybe also the opposite: Every unitary can be written as an exponential of an antihermitian operator).
- ☐ Poles and residues of complex functions
- ☐ Fourier transform
- ☐ exponential form of the sin and cos functions
- ☐ Euler's formula for complex exponentials
- ☐ Partial derivatives with respect to dependent variables (In this case: the parameters Pn and their complex conjugates Pn⍰)

- Relativity
  - ☒ Levi-Civita symbol
  - ☐ Roots of a 3rd order polynomial
- Coupled cluster
  - ☐ Taylor expansion
- Molecular properties
  - ☐ Writing cross product in terms of Levi-Civita symbols
  - ☐ Evaluating commutator of momentum operator with a local operator (like vector potential)
  - ☐ Lagrange multiplier method
  - ☐ Total and partial derivatives
- Multiconfigurational methods
  - ☐ resolution of the identity, projection operators

Note

Current levels and recommendation values coded in automatic generation of list script:

```
vels = ["beginner", "intermediate", "advanced"]

recommendations = ["math", "topic"]

# Topics and their descriptions, the latter used for writing the recomm
# exercises per topic.
topics = {
```

```
    "coupled-cluster": "Coupled-cluster theory",
    "second-quantization": "Second quantization",
    "scf": "Self-consistent field theory",
    "dft": "Density functional theory",
    "response-theory": "Response theory",
    "relativity": "Relativistic quantum chemistry",
    "molecular-properties": "Molecular properties",
    "multiconfig-methods": "Multiconfigurational methods",
}
```

Part I

Exercise recommendations

Recommended exercises for the first day

This chapter contains recommended exercises for the mathematics tutorial on the first day of the school. These recommended exercises are considered especially useful.

We provide also a list of recommended exercises for the 'beginners'. If you feel that you need training in the *absolute basics* of the mathematics used in quantum chemistry, we recommend that you start here.

In the next chapter, we provide lists of recommended exercises for each quantum chemistry topic taught at the school.

Beginner

- Exercise 1.1 — Days of the week
- Exercise 1.10 — Classically allowed region for Morse potential
- Exercise 3.1 — Very basic sums
- Exercise 3.2 — Simplifying as a sum
- Exercise 3.4 — Reconstruct a vector
- Exercise 3.5 — Summing in a matrix–vector product
- Exercise 3.6 — The summation index is a dummy index
- Exercise 3.7 — Linearity of sums
- Exercise 3.8 — When something does not depend on the summation index
- Exercise 3.9 — Free and summed indices
- Exercise 3.10 — Shifting the summation index
- Exercise 3.11 — The Kronecker delta: basic exercise
- Exercise 4.1 — Basic factorials
- Exercise 5.1 — Basic binomial coefficients
- Exercise 5.2 — Pascal's triangle
- Exercise 6.1 — Basic vector operations
- Exercise 6.2 — Matrix-vector products
- Exercise 6.3 — Matrix-matrix products
- Exercise 6.4 — Linear independence and basis

- Exercise 6.5 — Gaussian elimination and vector decomposition
- Exercise 7.1 — Cooking with vectors 1
- Exercise 10.1 — Bra-ket notation and column vectors

Intermediate

- Exercise 1.2 — Five first natural numbers
- Exercise 1.8 — De Morgan's laws
- Exercise 1.11 — Molecules as sets
- Exercise 2.1 — Countability of the integers
- Exercise 3.3 — Estimating $\sqrt{e}$ via a Taylor series
- Exercise 4.2 — Lunch line
- Exercise 6.6 — Office coordinates
- Exercise 6.7 — Boat sailing on the ocean
- Exercise 6.8 — An exercise by Robert Beezer
- Exercise 6.9 — Compute the action of a matrix on vectors
- Exercise 7.2 — Cooking with vectors 2
- Exercise 9.1 — Gaussian elimination and overlap matrix
- Exercise 14.1
- Exercise 14.2
- Exercise 15.1
- Exercise 17.1
- Exercise 17.4
- Exercise 18.1
- Exercise 18.4
- Exercise 20.1
- Exercise 20.4
- Exercise 20.6
- Exercise 21.1
- Exercise 24.1

For the curious

- Exercise 4.4 — Stirling's approximation
- Exercise 5.3 — Pascal's identity
- Exercise 5.4 — Pascal's triangle and the Sierpinski triangle
- Exercise 8.1 — Complex numbers as a real vector space

- Exercise 11.1
- Exercise 12.1
- Exercise 13.1
- Exercise 25.1
- Exercise 27.1

Recommended exercises by topic

This chapter contains lists of mathematics exercises grouped by each quantum chemistry topic taught at the school. These exercises can be useful as a warm-up to the proper exercises given each day.

Coupled-cluster theory

No exercises selected.

Second quantization

Beginner

- Exercise 3.1 — Very basic sums
- Exercise 3.2 — Simplifying as a sum
- Exercise 3.4 — Reconstruct a vector
- Exercise 3.6 — The summation index is a dummy index
- Exercise 3.7 — Linearity of sums
- Exercise 3.8 — When something does not depend on the summation index
- Exercise 3.9 — Free and summed indices
- Exercise 3.10 — Shifting the summation index
- Exercise 3.11 — The Kronecker delta: basic exercise
- Exercise 5.1 — Basic binomial coefficients

Self-consistent field theory

No exercises selected.

Density functional theory

No exercises selected.

Response theory

No exercises selected.

Relativistic quantum chemistry

Beginner

- Exercise 3.12 — The Levi–Civita symbol: basic exercise

Molecular properties

No exercises selected.

Multiconfigurational methods

No exercises selected.

Part II

Mathematical foundations

1 Sets and functions

Sets and set notation appear many places in quantum chemistry. The goal of this chapter is to train you in the basic usage of sets and set notation.

1.1 Exercises: abstract sets and functions

Exercise 1.1 (Days of the week)**.** Write the days of the week as a set

Exercise 1.2 (Five first natural numbers)**.** Write the five first natural numbers using set notation as indicated in the lecture notes

Exercise 1.3 (Ordered pair)**.** An *ordered pair* (x, y) is a two-element "set" where the order matters. Two ordered pairs (x, y) and (u, v) are equal if and only if $x = u$ and $y = v$. From this it follows that $(x, y) = (y, x)$ if and only if $x = y$. A definition of an ordered pair as a *set* is

$$(x, y) = \{\{x\}, \{x, y\}\}.$$

Show that $(x, y) = (y, x)$ if and only if $x = y$ using the set theoretic definition.

Exercise 1.4 (Rational numbers as ordered pairs)**.** A nonnegative rational number p/q, $p, q \in \mathbb{N}$, $q > 0$, can be written as an ordered pair (p, q). Write down the rational numbers 1/4, 2/3, 0, 1/3 as ordered pairs using set theory, both for the pair and for each integer. (Strictly speaking we the number itself is an *equivalence class* of such pairs, i.e., $pn/qn = p/q$, so we must identify (p, q) and (np, nq). Ignore equivalence classes in this exercise.)

Exercise 1.5 (Cartesian product of two sets)**.** The *cartesian product* of two sets A and B is

$$A \times B = \{(a, b) \mid a \in A,\ b \in B\}.$$

Write down the cartesian product of $\{\heartsuit, \diamondsuit, \spadesuit, \clubsuit\}$ and $\{1, 2, 3\}$, using the set theoretic definition for the ordered pair. You can skip spelling out the set theoretic definition of the natural numbers.

Exercise 1.6 (Ordered pair as a two-element set)**.** Similar to the previous exercise, write down $\{1, 2, 3\} \times \{2, 3, 4\}$.

Exercise 1.7 (Function from one set to another set)**.** A *function* $f : A \to B$ from one set A to another set B is a rule that assigns to every $a \in A$ precisely one $b \in B$. In terms of set theory, a function $f : A \to B$ is a subset of $A \times B$, such that

- For all $a \in A$ there exists $b \in B$ such that $(a, b) \in f$.
- For all $a \in A$ and $b, b' \in B$, if $(a, b) \in f$ and $(a, b') \in f$, then $b = b'$.

Write down the function $f : \{1, 2, 3\} \to \{1, 2, 3\}$, $f(1) = 2$, $f(2) = 3$, $f(3) = 1$, using the set theoretic definition of ordered pairs and the cartesian product.

De Morgan's laws are useful when discussing subsets A, B of a larger set X. Recall that the complement of A relative to X is $A^{\complement} = X \setminus A$. De Morgan's laws state that:

- $(A \cup B)^{\complement} = A^{\complement} \cap B^{\complement}$.
- $(A \cap B)^{\complement} = A^{\complement} \cup B^{\complement}$.

Exercise 1.8 (De Morgan's laws)**.** Draw a picture, representing X as the whole sheet, A and B as overlapping shapes, e.g., circles. Label A, B, $A \cap B$, and $A \cup B$, making another drawing if necessary. Convince yourself that De Morgan's laws are correct.

Exercise 1.9 (De Morgan's laws 2)**.** Prove De Morgan's laws mathematically. Note that the complement operation acts like negation of truth, i.e., $a \in A^{\complement}$ if and only if $a \in X$ and $a \notin A$.

1.2 Exercises: Sets in quantum chemistry

Exercise 1.10 (Classically allowed region for Morse potential)**.** Figure Figure 1.1 shows the *Morse potential* - a common model potential for vibrational motion of a diatomic molecule:

$$V(r) = D_e \left(1 - e^{-a(r - r_e)}\right)^2 - D_e,$$

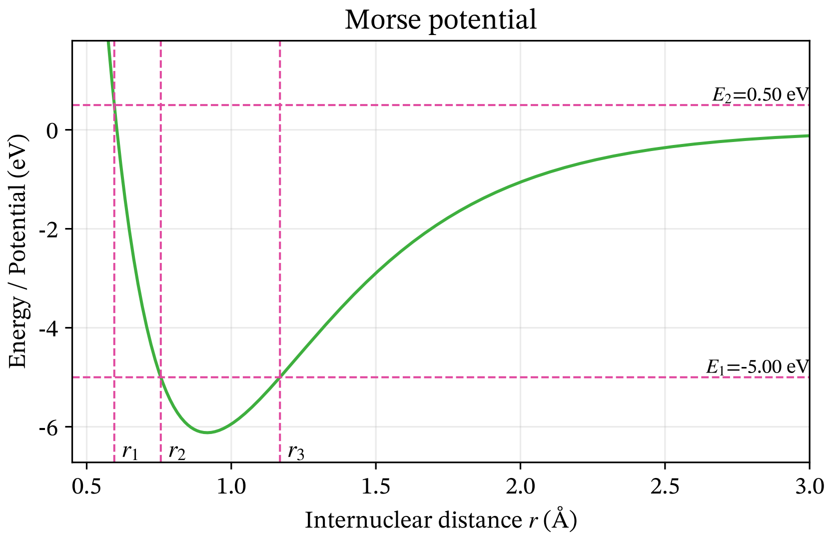


Figure 1.1: Morse potential

where $D_e > 0$ is the dissociation energy, $r_e > 0$ is the equilibrium bond length, and $a > 0$ is a parameter that determines the width of the potential well. The total energy of the molecule is denoted by $E \in \mathbb{R}$. The classically allowed region for the total energy E is the set S_E of all bond lengths r such that $V(r) \leq E$:

$$S_E = \{r \in \mathbb{R} \mid V(r) \leq E\}$$

a. Explain the notation $S_E = \{r \in \mathbb{R} \mid V(r) \leq E\}$ in words. In particular, what does the vertical bar $\mid$ mean?

b. Can you find a condition on E such that S_E is the empty set?

c. Can you find a condition on E such that S_E is the whole real line?

d. Based on the figure, which of the following statements are true?

- $S_{E_1} = (r_2, r_3)$
- $S_{E_1} = [r_2, r_3]$
- $S_{E_1} = \{r_2, r_3\}$

e. Based on the figure, which of the following statements are true?

- $S_{E_2} = (r_1, +\infty)$
- $S_{E_2} = [r_1, +\infty)$
- $S_{E_2} = \{r_1, r_2, r_3\}$

Exercise 1.11 (Molecules as sets)**.** In this exercise, a *molecule* is informally defined only by its chemical composition: the number of atoms of each element occurring in its molecular formula. Thus, H_2 and CO are two distinct molecules, while HCOOH and CO_2H_2 represent the same molecule, since both contain one C atom, two H atoms, and two O atoms.

We ignore molecular geometry, bonding, isotopes, charge, and all questions of chemical stability. In particular, any formal combination of atoms is deemed a molecule, whether or not such a species could exist physically.

We restrict our attention to the four elements C, H, O, and N. Let M denote the set of all molecules that can be constructed from these elements. A molecule must contain at least one atom.

One useful way of describing a molecule is by the ordered quadruple

$$(n_{\mathrm{C}}, n_{\mathrm{H}}, n_{\mathrm{O}}, n_{\mathrm{N}}), \tag{1.1}$$

where each n_X is the number of atoms of element X in the molecule. For example,

$$\mathrm{CH_4} \longleftrightarrow (1, 4, 0, 0),$$

and

$$\mathrm{N_2O} \longleftrightarrow (0, 0, 1, 2).$$

a. Explain why Equation 1.1 uniquely determines an element of M. Give 4-5 additional examples of molecules in terms of this representation.

b. Express M in terms of $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. What is the cardinality of M? Is it finite, countably infinite, or uncountably infinite? Justify your answer.

c. Let

$$M_2 = \{m \in M \mid m \text{ contains exactly two atoms}\}.$$

Find $|M_2|$ and list all elements of M_2.

d. Let $S \subset M$ be the set

$$S = \{m \in M \mid m \text{ contains at least one H atom}\}.$$

Describe S in terms of the quadruple representation above. Is S finite, countably infinite, or uncountably infinite? Justify your answer.

   d. Describe the complement $M \setminus S$. What is its cardinality?

e. Define

$$C = \{m \in M \mid m \text{ contains at least one C atom}\}.$$

Describe the following sets both in words and in terms of the quadruple representation:

$$C \cap S, \qquad C \cup S, \qquad C \setminus S.$$

Determine the cardinality of each set.

f. For $n \geq 1$, define

$$M_n = \{m \in M \mid m \text{ contains exactly } n \text{ atoms}\}.$$

Thus, a molecule in M_n satisfies

$$n_{\mathrm{C}} + n_{\mathrm{H}} + n_{\mathrm{O}} + n_{\mathrm{N}} = n.$$

Show that

$$|M_n| = \binom{n+3}{3}.$$

Check that your formula agrees with your answer to part **b**.

g. Explain why

$$M = \bigcup_{n=1}^{\infty} M_n.$$

Are the sets M_n pairwise disjoint? Use this decomposition to give another argument that M is countably infinite.

h. Show that M and S have the same cardinality. Explain why this is possible even though S is a proper subset of M.

1.3 Solutions

1.3.1 Tutor note: Sets etc.

The first two exercises should be accessible to everyone. The von Neumann representation of natural numbers grows quickly, so five numbers suffice.

The visual exercise on De Morgan's laws supports the later discussion of open sets and is recommended.

Solution to Exercise 1.1

$$U = \{\text{Mandag, Tirsdag, Onsdag, Torsdag, Fredag, Lørdag, Søndag}\}$$

Solution to Exercise 1.2

Okay, so each number doubles in size. After ten numbers, we have on the order of 1000 characters per number. So the five first is enough.

With $0 = \varnothing$ and $n + 1 = n \cup \{n\}$, the first five natural numbers are

$$
\begin{aligned}
0 &= \varnothing, \\
1 &= \{\varnothing\}, \\
2 &= \{\varnothing, \{\varnothing\}\}, \\
3 &= \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\}, \\
4 &= \{0, 1, 2, 3\}.
\end{aligned}
$$

Solution to Exercise 1.3

The source does not supply a solution.

Solution to Exercise 1.4

The source does not supply a solution.

Solution to Exercise 1.5

The source does not supply a solution.

Solution to Exercise 1.6

The source does not supply a solution.

Solution to Exercise 1.7

The source does not supply a solution.

Solution to Exercise 1.8

The source does not supply a solution.

Solution to Exercise 1.9

Here is the proof of the first one: $x \in (A \cup B)^{\complement}$ iff $x \notin A \cup B$ iff $x \notin A$ and $x \notin B$ iff $x \in A^{\complement}$ and $x \in B^{\complement}$ iff $x \in A^{\complement} \cap B^{\complement}$.

Solution to Exercise 1.10

a. The notation

$$S_E = \{r \in \mathbb{R} \mid V(r) \leq E\}$$

means that S_E is the set of all real numbers r for which the potential energy $V(r)$ is less than or equal to the total energy E.

The vertical bar | means **such that**. Thus, the expression can be read as

> the set of all r in the real numbers such that $V(r) \leq E$.

b. The Morse potential has its minimum at the equilibrium bond length $r = r_e$. At this point,

$$V(r_e) = D_e(1 - e^0)^2 - D_e = -D_e.$$

Therefore,

$$V(r) \geq -D_e$$

for all r. If

$$E < -D_e,$$

there are no values of r for which $V(r) \leq E$. Hence,

$$\boxed{S_E = \varnothing \quad \text{if} \quad E < -D_e.}$$

Notice that when $E = -D_e$, the set is not empty; it contains exactly the equilibrium bond length:

$$S_{-D_e} = \{r_e\}.$$

c. There is no finite value of E for which S_E is the whole real line.

To see this, consider the limit as $r \to -\infty$. Since $a > 0$,

$$e^{-a(r-r_e)} \to +\infty,$$

and therefore

$$V(r) \to +\infty.$$

Thus, for any finite $E \in \mathbb{R}$, there are sufficiently negative values of r for which

$$V(r) > E.$$

Consequently,

$$\boxed{S_E \neq \mathbb{R} \quad \text{for every finite } E \in \mathbb{R}.}$$

d. At energy E_1, the horizontal energy line intersects the potential at the two turning points r_2 and r_3. Between these points,

$$V(r) \leq E_1.$$

The endpoints themselves are included because

$$V(r_2) = V(r_3) = E_1$$

and the definition uses $\leq$, rather than $<$.

Therefore,

$$\boxed{S_{E_1} = [r_2, r_3].}$$

Hence the statements have truth values

- $S_{E_1} = (r_2, r_3)$: **false**

- $S_{E_1} = [r_2, r_3]$: **true**
- $S_{E_1} = \{r_2, r_3\}$: **false**

e. At energy E_2, the potential first becomes less than or equal to E_2 at $r = r_1$. For all larger r, the potential remains below or equal to E_2.

Again, the point r_1 itself is included because

$$V(r_1) = E_2.$$

Therefore,

$$\boxed{S_{E_2} = [r_1, +\infty).}$$

Hence the statements have truth values

- $S_{E_2} = (r_1, +\infty)$: **false**
- $S_{E_2} = [r_1, +\infty)$: **true**
- $S_{E_2} = \{r_1, r_2, r_3\}$: **false**

Solution to Exercise 1.11

a. A molecule is uniquely determined by an ordered quadruple

$$(n_{\mathrm{C}}, n_{\mathrm{H}}, n_{\mathrm{O}}, n_{\mathrm{N}}),$$

where each entry is a non-negative integer and not all four entries are zero. Hence

$$M = \mathbb{N}_0^4 \setminus \{(0, 0, 0, 0)\}.$$

Since $\mathbb{N}_0$ is countably infinite, any finite Cartesian product of copies of $\mathbb{N}_0$ is also countably infinite. Removing one element does not change the cardinality. Therefore

$$|M| = \aleph_0,$$

so M is **countably infinite**.

b. A diatomic molecule corresponds to a non-negative integer solution of

$$n_{\mathrm{C}} + n_{\mathrm{H}} + n_{\mathrm{O}} + n_{\mathrm{N}} = 2.$$

The possibilities are

$$\mathrm{C_2}, \quad \mathrm{H_2}, \quad \mathrm{O_2}, \quad \mathrm{N_2}, \quad \mathrm{CH}, \quad \mathrm{CO}, \quad \mathrm{CN}, \quad \mathrm{HO}, \quad \mathrm{HN}, \quad \mathrm{ON}.$$

Hence

$$|M_2| = 10.$$

c. In the quadruple representation,

$$S = \left\{(n_{\mathrm{C}}, n_{\mathrm{H}}, n_{\mathrm{O}}, n_{\mathrm{N}}) \in \mathbb{N}_0^4 \mid n_{\mathrm{H}} \geq 1\right\}.$$

The set S is infinite, since it contains

$$\mathrm{H}, \mathrm{H_2}, \mathrm{H_3}, \ldots$$

It is also a subset of the countable set M. Therefore

$$|S| = \aleph_0,$$

so S is countably infinite.

d. The complement consists of all molecules containing no hydrogen:

$$M \setminus S = \{(n_{\mathrm{C}}, 0, n_{\mathrm{O}}, n_{\mathrm{N}}) \mid n_{\mathrm{C}}, n_{\mathrm{O}}, n_{\mathrm{N}} \in \mathbb{N}_0,\ n_{\mathrm{C}} + n_{\mathrm{O}} + n_{\mathrm{N}} > 0\}.$$

This set is again countably infinite:

$$|M \setminus S| = \aleph_0.$$

e. We have

$$C = \{(n_\mathrm{C}, n_\mathrm{H}, n_\mathrm{O}, n_\mathrm{N}) \in M \mid n_\mathrm{C} \geq 1\}.$$

The intersection

$$C \cap S = \{(n_\mathrm{C}, n_\mathrm{H}, n_\mathrm{O}, n_\mathrm{N}) \in M \mid n_\mathrm{C} \geq 1,\ n_\mathrm{H} \geq 1\}$$

consists of molecules containing both carbon and hydrogen.

The union

$$C \cup S = \{(n_\mathrm{C}, n_\mathrm{H}, n_\mathrm{O}, n_\mathrm{N}) \in M \mid n_\mathrm{C} \geq 1 \text{ or } n_\mathrm{H} \geq 1\}$$

consists of molecules containing carbon or hydrogen, or both.

Finally,

$$C \setminus S = \{(n_\mathrm{C}, 0, n_\mathrm{O}, n_\mathrm{N}) \in M \mid n_\mathrm{C} \geq 1\}$$

consists of molecules containing carbon but no hydrogen.

All three sets are infinite subsets of the countable set M, and each is easily seen to contain a countably infinite sequence of distinct molecules. Hence

$$|C \cap S| = |C \cup S| = |C \setminus S| = \aleph_0.$$

f. The number of elements of M_n is the number of non-negative integer solutions of

$$n_\mathrm{C} + n_\mathrm{H} + n_\mathrm{O} + n_\mathrm{N} = n.$$

By the stars-and-bars formula, the number of such solutions is

$$|M_n| = \binom{n+4-1}{4-1} = \binom{n+3}{3}.$$

For $n = 2$,

$$|M_2| = \binom{5}{3} = 10,$$

in agreement with part **b**.

g. Every molecule contains some finite positive number n of atoms, so it belongs to exactly one set M_n. Therefore

$$M = \bigcup_{n=1}^{\infty} M_n.$$

The sets M_n are pairwise disjoint, since a molecule cannot contain exactly n atoms and exactly m atoms when $n \neq m$.

Each M_n is finite, since

$$|M_n| = \binom{n+3}{3}.$$

Thus M is a countable union of finite sets. Therefore M is countable. Since there are infinitely many molecules, M is countably infinite.

h. From parts **a** and **c**,

$$|M| = |S| = \aleph_0.$$

Thus M and S have the same cardinality, even though

$$S \subsetneq M.$$

This is possible because M is infinite. Unlike finite sets, an infinite set can have the same cardinality as one of its proper subsets.

For example, enumerate the molecules as

$$M = \{m_1, m_2, m_3, \ldots\}$$

and the hydrogen-containing molecules as

$$S = \{s_1, s_2, s_3, \dots\}.$$

Then the map

$$f : M \to S, \qquad f(m_i) = s_i,$$

is a bijection. Hence M and S have the same cardinality.

2 Cardinality and countability

Cardinality compares sets through bijections. The exercises cover countable sets, Cantor's diagonal argument, and the cardinality of the continuum.

Recall that the *cardinality* $|S|$ of a set S is the number of elements in S. For finite sets, this is intuitive, but what about infinite sets?

One says that A and B have the same cardinality if there exists a bijection $f : A \to B$, i.e., f is one-to-one and onto. (Intuitively, you can draw lines between individual elements of A and B, only one line to/from each element. Precisely: Onto means, to every $b \in B$ there exists a $a \in A$ such that $f(a) = b$. One-to-one: $f(a) = f(a')$ implies $a = a'$.) One can think of the bijection as labelling each $b \in B$ by exactly one element $a \in A$, and that all labels from A are used up.

Let $\mathbb{N}_n = \{1, 2, \cdots, n\}$. This set has cardinality n.

The set $\mathbb{N}$ has an infinite number of elements. By definition, $|\mathbb{N}| = \aleph_0$ ("aleph-nought").

2.1 Exercises

Exercise 2.1 (Countability of the integers)**.** Show that $|\mathbb{Z}| = \aleph_0$ by explicitly constructing a bijection $f : \mathbb{N} \to \mathbb{Z}$.

Exercise 2.2 (Countability of the natural number plane)**.** Show that $|\mathbb{N} \times \mathbb{N}| = \aleph_0$. Hint: Draw a picture of $\mathbb{N} \times \mathbb{N}$, and try to draw a line through all the points in this set. How does this show the existence of a bijection between $\mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$?

Exercise 2.3 (Countability of the rational numbers)**.** Show that $|\mathbb{Q}| = \aleph_0$.

Exercise 2.4 (Interval vs. real numbers)**.** Show that the interval $I = (-1, 1)$ has the same cardinality as $\mathbb{R}$. You must construct a function $f : (-1, 1) \to \mathbb{R}$ that is a bijection.

Exercise 2.5 (Uncountability of the real numbers)**.** In this exercise, we show that $|\mathbb{R}| > \aleph_0$. The cardinality would be $\aleph_0$ if we could write a list of the real numbers. It suffices to find a list of the numbers in $I = [0,1[$, i.e., infinite decimal expansions $0.d_1d_2d_3\cdots$, with $d_i \in \{0,1,2,3,4,5,6,7,8,9\}$.

We will look for a contradiction. Consider a list $x_1, x_2, x_3, \dots$ of all real numbers in $[0,1[$. This is essentially an infinite matrix of digits. Can you find a real number $x \in [0,1]$ which is *not* in this list, i.e., $x \neq x_j$ for all j? Hint: consider the diagonal of the matrix.

The cardinality of $\mathbb{R}$, "the continuum", is written $2^{\aleph_0}$.

Exercise 2.6 (Constructing a one-to-one map)**.** Show that $|[0,1[\times[0,1[| = |[0,1[|$, by constructing a one-to-one map. Hint: Use decimal expansions. Conclude that also $|\mathbb{R}^2| = |\mathbb{C}| = |\mathbb{R}|$.

2.2 Solutions

Solution to Exercise 2.1

We construct a map $f : \mathbb{N} \to \mathbb{Z}$. We let the even numbers in $\mathbb{N}$ map to positive numbers in $\mathbb{Z}$, and odd numbers in $\mathbb{N}$ to negative numbers. For every $n \in \mathbb{N}$,

$$f(2n) = n, \quad f(2n+1) = -n-1.$$

We get $f(0) = 0$, $f(1) = -1$, $f(2) = 1$, $f(3) = -2$, etc.

Solution to Exercise 2.2

Draw a picture of $\mathbb{N}\times\mathbb{N}$ - a uniform grid. The line can be drawn in many ways, but the easiest is perhaps to connect $(0,0)-(0,1)-(1,0)-(2,0)-(1,1)-(0,2)-\dots$

Solution to Exercise 2.3

Use that $x \in \mathbb{Q}$ is represented as a fraction, i.e., a point in $\mathbb{N}\times\mathbb{N}$. Actually, x is an equivalence class, so some points in $\mathbb{N}\times\mathbb{N}$ are eliminated.

Solution to Exercise 2.4

One possible bijection is

$$f(x) = \frac{x}{1-|x|}, \qquad -1 < x < 1.$$

Its inverse is $g(y) = y/(1+|y|)$ for $y \in \mathbb{R}$.

Solution to Exercise 2.5

The proof is called Cantor's diagonalization argument, and is used in several other important results. (For example, to prove that there is no Turing machine that can determine whether an arbitrary Turing machine stops.)

Let the kth digit be different from the kth digit of number k in the list. Then the real number thus constructed cannot be equal to any of the numbers in the list.

Solution to Exercise 2.6

Let every even-numbered decimal from x be the decimal of y_1 and odd decimal the decimal of y_2, respectively. This maps decimal expansions x onto pairs of decimal expansions (y_1, y_2). Clearly, the inverse mapping can be created, too.

3 Summing over indices

Summing several things in a single expression occurs frequently in mathematics, and in particular in applications to physics and chemistry. The general syntax for a summation over an index is:

$$\sum_{i \in I} f(i),$$

where I is some countable set of indices, i is the index of summation, and $f(i)$ is the function/quantity being summed. The set I can be finite or infinite, and the index i can be a number, a vector, or even a more complicated object.

The most basic and common case is a sum over a finite or infinite set of integers, which is usually written as

$$\sum_{i=a}^{b} f(i)$$

where i is the index of summation, a is the lower limit of the index i, b is the upper limit, and $f(i)$ is the function/quantity being summed.

Here are some concrete examples:

1. A sum over a finite number of terms:

$$\sum_{i=1}^{5} i = 1 + 2 + 3 + 4 + 5$$

2. An infinite series:

$$\sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots$$

3. A series expansion. In this example, the exponential function is expressed as a Taylor series, summing over all non-negative integers:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \ldots$$

4. Vector notation. For example, a vector $\mathbf{r} \in \mathbb{R}^3$ can be expressed as a sum over its components:
$$\sum_{i=1}^{3} r_i \mathbf{e}_i = \mathbf{r}$$

5. A matrix-vector product expressed as a sum over indices: The i-th component of the product of a matrix A and a vector $\mathbf{x}$ is given by the sum over the products of the corresponding elements of the matrix and vector:
$$(A\mathbf{x})_i = \sum_{j=1}^{n} A_{ij} x_j$$

3.0.1 Manipulating sums

The index in a sum is called a **summation index** or **dummy index**. Its name has no significance. For example,

$$\sum_{i=1}^{n} a_i = \sum_{j=1}^{n} a_j.$$

The index may therefore be renamed, as long as it is renamed consistently within the sum.

Summation is **linear**. Sums can be split or combined,

$$\sum_{i=1}^{n} (a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i,$$

and factors that do not depend on the summation index can be taken outside the sum,

$$\sum_{i=1}^{n} c a_i = c \sum_{i=1}^{n} a_i,$$

where it is important that c is independent of i.

Expressions may contain both summed and unsummed indices. An index that is not summed over is called a **free index**. For example, in the matrix-vector product

$$y_i = \sum_{j=1}^{n} A_{ij} x_j,$$

j is a dummy index, while i is a free index. The equation therefore represents one equation for each possible value of i.

3.0.2 Special symbols used in summation notation

The **Kronecker delta** δ_{ij} is a special symbol that is often used in summation notation. It is defined as

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Sometimes, if confusion can arise one inserts a comma between the two indices, writing $\delta_{i,j}$ instead of δ_{ij}.

The Kronecker delta forms the matrix elements of the identity matrix.

The **Levi-Civita symbol** ϵ_{ijk} is another special symbol that is often used in summation notation in the three-dimensional case, i.e., where $i, j, k \in \{1, 2, 3\}$. It is defined as

$$\epsilon_{ijk} = \begin{cases} 1 & \text{if } (i, j, k) \text{ is an even permutation of } (1, 2, 3), \\ -1 & \text{if } (i, j, k) \text{ is an odd permutation of } (1, 2, 3), \\ 0 & \text{if any two indices are equal.} \end{cases}$$

3.1 Exercises: Basic summation

These exercises are at a very basic level. If you are already comfortable with summation notation, you may skip them.

Exercise 3.1 (Very basic sums). Expand the following sums and calculate their values:

a) $\sum_{i=1}^{4} i$

b) $\sum_{i=1}^{4} i^2$

c) $\sum_{k=0}^{3} 2^k$

Exercise 3.2 (Simplifying as a sum)**.** Write each expression using summation notation:

a) $x_1 + x_2 + x_3 + x_4 + x_5$

b) $1 + q + q^2 + q^3 + q^4$

c) $a_1b_1 + a_2b_2 + a_3b_3$

Exercise 3.3 (Estimating $\sqrt{e}$ via a Taylor series)**.** Use the first four terms of the series for e^x to estimate $\sqrt{e}$:

$$\sqrt{e} = e^{0.5} \approx \sum_{n=0}^{3} \frac{(0.5)^n}{n!}.$$

Write out the four terms and add them. The estimate is not especially accurate; the point is to see how the index n controls both the power and the factorial.

Exercise 3.4 (Reconstruct a vector)**.** The components of a vector are $r_1 = 2$, $r_2 = -1$, and $r_3 = 4$. Let $\mathbf{e}_i$ be the standard basis in $\mathbb{R}^3$.

a) Write out $\sum_{i=1}^{3} r_i\mathbf{e}_i$ term by term.

b) Gather the result as a column vector

Exercise 3.5 (Summing in a matrix–vector product)**.** Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 0 & 3 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

Use

$$(A\mathbf{x})_i = \sum_{j=1}^{3} A_{ij}x_j$$

to write out and calculate $(A\mathbf{x})_1$ and $(A\mathbf{x})_2$. Collect the two answers in a column vector.

Exercise 3.6 (The summation index is a dummy index)**.** Consider

$$S = \sum_{i=1}^{4} i^2.$$

a. Write out the sum explicitly and calculate its value.

b. Write out and calculate

$$\sum_{j=1}^{4} j^2.$$

c. Explain why

$$\sum_{i=1}^{4} i^2 = \sum_{j=1}^{4} j^2.$$

d. Rewrite

$$\sum_{k=1}^{5} a_k b_k$$

using p as the summation index.

Exercise 3.7 (Linearity of sums)**.** Let a_i and b_i, $i = 1, 2, 3$ be two sets of numbers and let c be a constant, i.e., just a number.

a. Write out both sides of

$$\sum_{i=1}^{3}(a_i + b_i) = \sum_{i=1}^{3} a_i + \sum_{i=1}^{3} b_i$$

and verify that they are equal.

b. Write out both sides of

$$\sum_{i=1}^{3} ca_i = c\sum_{i=1}^{3} a_i$$

and verify that they are equal.

c. Use these properties to rewrite

$$\sum_{i=1}^{3}(2a_i - 3b_i)$$

as two separate sums (i.e., each has a single summation sign).

Exercise 3.8 (When something does not depend on the summation index)**.** Let x be a number independent of the summation index i.

a. Simplify
$$\sum_{i=1}^{5} x.$$

b. Simplify
$$\sum_{i=1}^{5} ix.$$

c. Explain why x can be taken outside the sum in
$$\sum_{i=1}^{n} ix = x\sum_{i=1}^{n} i.$$

Exercise 3.9 (Free and summed indices)**.** Consider the following expression for the i-th component of a vector $\mathbf{y}$, being the product of a matrix A and a vector $\mathbf{x}$:
$$y_i = \sum_{j=1}^{3} A_{ij}x_j.$$

a. Which index is summed over?

b. Which index is not summed over?

c. For $i = 2$, write the right-hand side explicitly.

d. If A has four rows and three columns, how many components does $\mathbf{y}$ have?

e. Explain why replacing j by k,
$$y_i = \sum_{k=1}^{3} A_{ik}x_k,$$
does not change the equation.

Exercise 3.10 (Shifting the summation index)**.** Consider
$$S = \sum_{i=1}^{4} a_i.$$

a. Write out S explicitly.

b. Show that the same sum can be written as

$$S = \sum_{j=0}^{3} a_{j+1}$$

by writing out the terms.

c. Rewrite

$$\sum_{i=2}^{5} a_i$$

as a sum whose index starts at $j = 0$.

d. Rewrite

$$\sum_{i=0}^{n-1} a_{i+1}$$

as a sum with index j running from 1 to n.

Exercise 3.11 (The Kronecker delta: basic exercise). The **Kronecker delta** is defined in the introduction to this chapter. It is a special symbol that is often used in summation notation.

a. Evaluate δ_{11}, δ_{12}, δ_{23}, and δ_{33}.

b. Let $a_1 = 2$, $a_2 = 5$, and $a_3 = 7$. Write out all terms in the sum

$$\sum_{j=1}^{3} \delta_{2,j} a_j$$

and evaluate it.

c. Repeat (b) for

$$\sum_{j=1}^{3} \delta_{1j} a_j \qquad \text{and} \qquad \sum_{j=1}^{3} \delta_{3j} a_j.$$

d. Based on your results, simplify the general expression

$$\sum_{j=1}^{n} \delta_{ij} a_j.$$

e. In the expression in (d), which index is a dummy index and which is a free index? What values can the free index take?

Exercise 3.12 (The Levi–Civita symbol: basic exercise)**.** The **Levi–Civita symbol** ε_{ijk} is defined in the introduction to this chapter.

a. Evaluate ε_{123}, ε_{132}, ε_{231}, ε_{321}, and ε_{223}.

b. Find all triples (i, j, k) for which $\varepsilon_{ijk} = +1$.

c. Find all triples (i, j, k) for which $\varepsilon_{ijk} = -1$.

d. What happens to ε_{ijk} if two indices are interchanged? Illustrate with two examples.

e. Let a_1, a_2, and a_3 be the components of a vector. Write out the sum

$$\sum_{j=1}^{3}\sum_{k=1}^{3} \varepsilon_{1jk} a_j b_k$$

and remove all terms that vanish because of the Levi–Civita symbol.

Exercise 3.13 (A molecular orbital in an atomic-orbital basis)**.** In quantum chemistry, a molecular orbital $\psi_p(\mathbf{r})$ is commonly expanded in a finite set of atomic-orbital basis functions $\chi_\mu(\mathbf{r})$,

$$\psi_p(\mathbf{r}) = \sum_{\mu=1}^{K} C_{\mu p} \chi_\mu(\mathbf{r}).$$

Here, μ labels the basis functions, while p labels the molecular orbital.

Consider three basis functions and the coefficient matrix

$$C = \begin{bmatrix} 0.7 & 0.5 \\ 0.7 & -0.5 \\ 0.1 & 0.7 \end{bmatrix}.$$

a. In the expression for $\psi_p(\mathbf{r})$, identify the dummy index and the free index.

b. Write $\psi_1(\mathbf{r})$ explicitly as a linear combination of $\chi_1(\mathbf{r})$, $\chi_2(\mathbf{r})$, and $\chi_3(\mathbf{r})$, using the matrix elements of C.

c. Do the same for $\psi_2(\mathbf{r})$.

d. Suppose that at a particular point $\mathbf{r}_0$,

$$\chi_1(\mathbf{r}_0) = 1.0, \qquad \chi_2(\mathbf{r}_0) = 0.5, \qquad \chi_3(\mathbf{r}_0) = 0.2.$$

Calculate $\psi_1(\mathbf{r}_0)$ and $\psi_2(\mathbf{r}_0)$.

e. Explain why the summation index μ could be replaced by another symbol, such as i, without changing the meaning of the expression.

4 Combinatorics

Combinatorics is the part of mathematics concerned with *counting* in a broad sense, for example counting the number of ways to arrange a set of objects or the number of combinations that can be made from a set of objects. It is a fundamental part of discrete mathematics and has applications in many areas of mathematics, computer science, chemistry, and physics.

4.1 Exercises: Factorials

The *factorial* of a non-negative integer n is defined as the product of all positive integers less than or equal to n:

$$n! = n \cdot (n-1) \cdot (n-2) \cdots 3 \cdot 2 \cdot 1, \quad n \geq 1.$$

It is also defined that $0! = 1$. The factorial appears in many mathematical contexts, including series expansions, and of course combinatorics. Its basic use is the following:

> **Factorial and counting**
>
> Given a set of n distinct objects, the number of ways to arrange them in a sequence (i.e., the number of permutations) is $n!$.

In one exercise, you will need the following approximation for large n, known as Stirling's approximation:

> **Stirling's approximation**
>
> For large n, the factorial $n!$ can be approximated as

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

In calculations, especially numerical calculations, it is often convenient to use the logarithm of the factorial. Using Stirling's approximation, we have

$$\log(n!) \approx \frac{1}{2}\log(2\pi n) + n\log(n) - n.$$

Exercise 4.1 (Basic factorials)**.** Compute the factorials 0!, 1!, 2!, 3!, 4!, and 5!.

Exercise 4.2 (Lunch line)**.** 42 people stand in a lunch line. How many different lunch line arrangements with 42 people are possible?

Exercise 4.3 (Chemist's shelf)**.** Compute the number of arrangements for the following scenarios:

a. A chemist has 10 different chemical samples. How many ways can the chemist arrange the samples on a shelf?
b. A chemist has 10 different chemical samples. 4 of the samples are liquids, and the other 6 are solids. How many ways can the chemist arrange the samples on a shelf if all the liquids must be together?

Exercise 4.4 (Stirling's approximation)**.** We study Stirling's approximation for the factorial.

a. Use Stirling's approximation to estimate 5!. Compare your result with the exact result

$$5! = 120.$$

b. Use Stirling's approximation to estimate 10!. Compare again with the exact number

$$10! = 3\,628\,800.$$

c. The relative error of an approximation A to an exact value x is

$$\frac{|A - x|}{x}.$$

Calculate the relative error of Stirling's approximation for 5! and 10!. For which value of n is the approximation better?

d. Use Stirling's approximation to estimate 50!. The exact value is approximately

$$50! \approx 3.041 \times 10^{64}.$$

e. Based on your results above, what happens to the accuracy of Stirling's approximation as n increases?

5 Exercises: Binomial coefficients and Pascal's triangle

The *binomial coefficient* $\binom{n}{k}$ is defined as the number of ways to choose k elements from a set of n elements, without regard to the order of selection. It is given by the formula:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!},$$

Pascal's triangle is a triangular array of numbers where each number is the sum of the two numbers directly above it:

$$1 \tag{5.1}$$
$$1 \quad 1 \tag{5.2}$$
$$1 \quad 2 \quad 1 \tag{5.3}$$
$$1 \quad 3 \quad 3 \quad 1 \tag{5.4}$$
$$1 \quad 4 \quad 6 \quad 4 \quad 1 \tag{5.5}$$
$$\vdots \tag{5.6}$$

Fact: The n-th row of Pascal's triangle contains the binomial coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$.

Exercise 5.1 (Basic binomial coefficients)**.**

a. Compute the binomial coefficients $\binom{5}{2}$, $\binom{6}{3}$, and $\binom{7}{4}$.
b. In Norwegian Lotto, 7 numbers are drawn from a set of 34 numbers. How many different combinations of 7 numbers can be drawn?

Exercise 5.2 (Pascal's triangle)**.** Compute the first 6 rows of Pascal's triangle. Verify that the numbers in the n-th row correspond to the binomial coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$.

Exercise 5.3 (Pascal's identity)**.** The binomial coefficients satisfy **Pascal's identity**,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

a. Explain why this relation is equivalent to the construction of Pascal's triangle, where each number is the sum of the two numbers directly above it.

b. Write each binomial coefficient on the right-hand side in terms of factorials.

c. Put the two terms over a common denominator and simplify the expression.

d. Show that the resulting expression is

$$\frac{n!}{k!(n-k)!},$$

and hence prove Pascal's identity.

e. Give a combinatorial explanation of Pascal's identity. *Hint:* Consider choosing k objects from a set of n objects. Pick one particular object and separate the choices into two cases: those that contain this object and those that do not.

Exercise 5.4 (Pascal's triangle and the Sierpinski triangle)**.** Write a computer program to generate the first 2^n rows of Pascal's triangle, where $n = 1, 2, \dots$. Create a $2^n \times 2^n$ lower triangular matrix T, where $T_{k,n} = 0$ if $\binom{n}{k}$ is even, and $T_{k,n} = 1$ if the coefficient is odd, for $k \le n$ (indexing from zero). Visualize the matrix. What pattern do you observe? This pattern is known as the Sierpinski triangle.

If you use Python and sympy, you can generate binomial factors with the function `sympy.binomial(n, k)`. Check for parity using the modulo operator `%`. For example, `sympy.binomial(n, k) % 2` will return `0` for even numbers and `1` for odd numbers.

5.1 Solutions

Solution for Exercise 4.1

Recall that $0! = 1$ and, for a positive integer n,

$$n! = n(n-1)\cdots 2 \cdot 1.$$

Therefore,

$$0! = 1, \qquad 1! = 1, \qquad 2! = 2, \qquad 3! = 6, \qquad 4! = 24, \qquad 5! = 120.$$

Solution for Exercise 4.2

There are 42 choices for the first person in the line, 41 choices for the second, and so on. Hence the number of possible arrangements is

$$42! = 42 \cdot 41 \cdots 2 \cdot 1.$$

Solution for Exercise 4.3

a. The 10 different chemical samples can be arranged in

$$10! = 3\,628\,800$$

different ways.

b. Since the 4 liquids must be together, we can first regard them as a single object. We then have this object together with the 6 solids, giving 7 objects to arrange. These can be arranged in 7! ways. For each such arrangement, the 4 liquids can be arranged among themselves in 4!. Therefore, the total number of arrangements is

$$7!4! = 120\,960.$$

Solution for Exercise 4.4

We use Stirling's approximation,

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

a. For $n = 5$,

$$5! \approx \sqrt{10\pi}\left(\frac{5}{e}\right)^5 \approx 118.0.$$

The exact value is

$$5! = 120.$$

Thus, Stirling's approximation is already quite close for $n = 5$.

b. For $n = 10$,

$$10! \approx \sqrt{20\pi}\left(\frac{10}{e}\right)^{10} \approx 3.599 \times 10^6.$$

The exact value is

$$10! = 3.6288 \times 10^6.$$

Again, the approximation is very close to the exact value.

c. For 5!, the relative error is

$$\frac{|118.0 - 120|}{120} \approx 0.0165,$$

or about

$$1.65\%.$$

For 10!, the relative error is

$$\frac{|3.599 \times 10^6 - 3.6288 \times 10^6|}{3.6288 \times 10^6} \approx 0.0083,$$

or about

$$0.83\%.$$

Thus, the approximation is better for $n = 10$ than for $n = 5$.

d. For $n = 50$,

$$50! \approx \sqrt{100\pi}\left(\frac{50}{e}\right)^{50} \approx 3.036 \times 10^{64}.$$

For comparison, the exact value is approximately

$$50! \approx 3.041 \times 10^{64}.$$

e. The relative error becomes smaller as n increases. Thus, Stirling's approximation becomes increasingly accurate for large n.

Solution for Exercise 5.1

a. The binomial coefficient is given by

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Thus,

$$\binom{5}{2} = \frac{5!}{2!3!} = \frac{5 \cdot 4}{2 \cdot 1} = 10,$$

$$\binom{6}{3} = \frac{6!}{3!3!} = \frac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1} = 20,$$

and

$$\binom{7}{4} = \frac{7!}{4!3!} = \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} = 35.$$

b. Since the order in which the 7 numbers are drawn does not matter, the number of possible combinations is

$$\binom{34}{7} = \frac{34!}{7!27!} = \frac{34 \cdot 33 \cdot 32 \cdot 31 \cdot 30 \cdot 29 \cdot 28}{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 5\,379\,616.$$

Thus, there are **5 379 616** different combinations of 7 numbers.

Solution for Exercise 5.3

a. The relation is equivalent to the construction of Pascal's triangle because each number in the triangle is the sum of the two numbers directly above it. The left term on the right-hand side corresponds to the number directly above and to the left, while the right term corresponds to the number directly above and to the right:

$$\binom{0}{0} \tag{5.7}$$

$$\binom{1}{0} \quad \binom{1}{1} \tag{5.8}$$

$$\binom{2}{0} \quad \binom{2}{1} \quad \binom{2}{2} \tag{5.9}$$

$$\binom{3}{0} \quad \binom{3}{1} \quad \binom{3}{2} \quad \binom{3}{3} \tag{5.10}$$

$$\vdots \tag{5.11}$$

b. Using

$$\binom{n}{k} = \frac{n!}{k!(n-k)!},$$

we have

$$\binom{n-1}{k-1} = \frac{(n-1)!}{(k-1)!(n-k)!}$$

and

$$\binom{n-1}{k} = \frac{(n-1)!}{k!(n-k-1)!}.$$

c. We write both terms with the common denominator

$$k!(n-k)!.$$

Thus,

$$\binom{n-1}{k-1} = \frac{k(n-1)!}{k!(n-k)!},$$

and

$$\binom{n-1}{k} = \frac{(n-k)(n-1)!}{k!(n-k)!}.$$

Therefore,

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \frac{k(n-1)! + (n-k)(n-1)!}{k!(n-k)!}.$$

d. Factoring out $(n-1)!$ gives

$$\frac{[k + (n-k)](n-1)!}{k!(n-k)!} = \frac{n(n-1)!}{k!(n-k)!}.$$

Since

$$n(n-1)! = n!,$$

we obtain

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \frac{n!}{k!(n-k)!} = \binom{n}{k}.$$

Hence,

$$\boxed{\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}}.$$

e. Now consider choosing k objects from a set of n objects. Select one particular object, say x.

Every choice of k objects falls into exactly one of two cases:

- The choice contains x. Then we must choose the remaining $k-1$ objects from the other $n-1$ objects. There are

$$\binom{n-1}{k-1}$$

such choices.

- The choice does not contain x. Then all k objects must be chosen from the other $n-1$ objects. There are

$$\binom{n-1}{k}$$

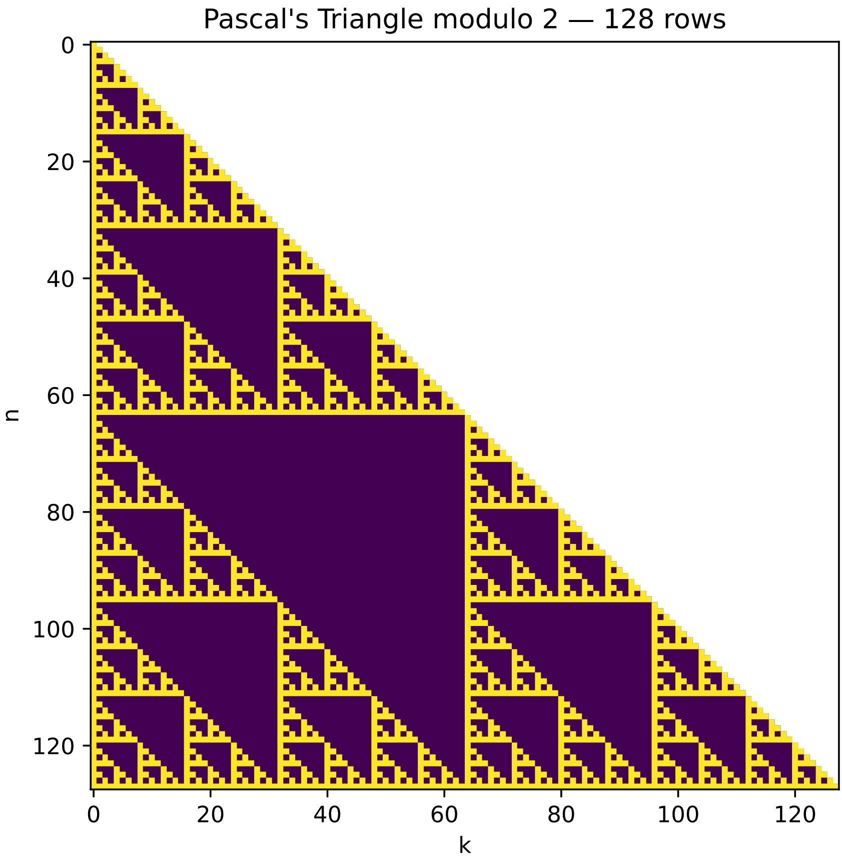
such choices.

Since these two cases are disjoint and include all possible choices,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Solution for Exercise 5.4

```python
from sympy import binomial
import numpy as np
import matplotlib.pyplot as plt

nrows = 128

# Build parity matrix: 1 = odd, 0 = even
T = np.zeros((nrows, nrows), dtype=int)

for n in range(nrows):
    for k in range(n + 1):
        T[n, k] = int(binomial(n, k) % 2)

# Plot only the valid triangular region cleanly
masked = np.ma.masked_where(
    np.fromfunction(lambda n, k: k > n, (nrows, nrows)),
    T
)

plt.figure(figsize=(6, 6))
plt.imshow(masked, origin="upper", interpolation="nearest", aspect="equ
plt.title(f"Pascal's Triangle modulo 2 - {nrows} rows")
plt.xlabel("k")
plt.ylabel("n")
plt.show()
```


Pascal's Triangle modulo 2 — 128 rows
n
k
0
20
40
60
80
100
120
0
20
40
60
80
100
120

Part III

Linear algebra

6 Vectors and matrices

The vector space $\mathbb{R}^n$ consists of all column vectors

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad x_i \in \mathbb{R},$$

with the usual vector addition and scalar multiplication. Similarly, $\mathbb{C}^n$ consists of column vectors with complex components. More generally, a vector space V over a field $\mathbb{F}$ is a set equipped with vector addition and scalar multiplication such that, for all $x, y, z \in V$ and $\alpha, \beta \in \mathbb{F}$,

1. **Commutativity of addition:** $x + y = y + x$.
2. **Associativity of addition:** $x + (y + z) = (x + y) + z$.
3. **Additive identity:** There exists a zero vector $0 \in V$ such that $x + 0 = x$.
4. **Additive inverse:** For every $x \in V$, there exists $-x \in V$ such that $x + (-x) = 0$.
5. **Distributivity over vector addition:** $\alpha(x + y) = \alpha x + \alpha y$.
6. **Distributivity over scalar addition:** $(\alpha + \beta)x = \alpha x + \beta x$.
7. **Associativity of scalar multiplication:** $(\alpha\beta)x = \alpha(\beta x)$.
8. **Scalar identity:** $1x = x$.

For a matrix $A \in \mathbb{R}^{m\times n}$ and a column vector $x \in \mathbb{R}^n$, the matrix-vector product $y = Ax \in \mathbb{R}^m$ is defined componentwise by

$$y_i = \sum_{j=1}^{n} A_{ij} x_j.$$

For matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, the matrix product $C = AB \in \mathbb{R}^{m \times p}$ is defined by

$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}.$$

The same definitions of matrix-vector and matrix-matrix multiplication apply over $\mathbb{C}$.

6.1 Exercises: Basic vectors and matrix calculations

Exercise 6.1 (Basic vector operations)**.** Consider the vectors

$$x = (1, 2, -1), \qquad y = (3, -1, 2), \qquad z = (0, 2, 1)$$

in the vector space $\mathbb{R}^3$.

a. Compute $x + y$.
b. Compute $x - y$.
c. Compute $2x$ and $-3y$.
d. Compute the linear combination $2x + 3y$.
e. Compute $x + 2y - z$.
f. Find numbers α and β such that $\alpha x + \beta y = (7, 0, 3)$.
g. Verify by direct calculation that $2(x + y) = 2x + 2y$. Which vector-space axiom does this illustrate?
h. Is $(1, 2)$ an element of this vector space? Is $(0, 0, 0)$ an element? Explain briefly.

Exercise 6.2 (Matrix-vector products)**.** Consider the matrices and vectors

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix}, \qquad x = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \qquad y = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}.$$

a. Compute Ax.

b. Compute By.

c. Compute $A(2x)$ and $2(Ax)$. Verify that they are equal.

d. Can the products Ay and Bx be formed? Explain your answer in terms of the dimensions of the matrices and vectors.

Exercise 6.3 (Matrix-matrix products)**.** Consider the matrices

$$A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ -1 & 4 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 2 \end{pmatrix}.$$

a. Compute AB.

b. Compute BA. Is $AB = BA$?

c. Compute AC. What are the dimensions of the resulting matrix?

d. Can the product CA be formed? Can the product BC be formed? Explain your answers using the dimensions of the matrices.

Exercise 6.4 (Linear independence and basis)**.** We work in the space $V = \mathbb{R}^2$, plane vectors. We define two vectors,

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

a. Show by elementary means that the two vectors are linearly independent.
b. Why do they form a basis for V?
c. Make a 2D drawing or plot of them as arrows from the origin.

Exercise 6.5 (Gaussian elimination and vector decomposition)**.** Let

$$\mathbf{v} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Using Gaussian elimination, compute the coefficients x_1 and x_2 such that

$$\mathbf{v} = x_1\mathbf{b}_1 + x_2\mathbf{b}_2.$$

Illustrate this decomposition in the graph from the previous exercise, using the parallelogram rule: The sum of two vectors is obtained by placing the origin of one at the end of the second, or vice versa.

Exercise 6.6 (Office coordinates)**.** A person walks from home. First, they walk to the subway station. This is 1 km north and 500 m east as the crow flies. The subway takes the person to the city centre, which 3 km east and 2 km south of the subway station, as the crow flies. The person then walks 200 m south and 400 m west to their office.

What are the coordinates of the office, relative to the person's home?

What is the distance to home, as the crow flies?

Exercise 6.7 (Boat sailing on the ocean)**.** A boat is sailing on the ocean, at a constant speed 20 km/h relative to the water surface, in the northeast direction. However, the water surface is moving too, at a constant velocity. After two hours, the boat is located 20 km north and 10 km east of the starting point. What is the speed and direction of ocean drift?

Exercise 6.8 (An exercise by Robert Beezer)**.** A three-digit number has two properties. The tens-digit and the ones-digit add up to 5. If the number is written with the digits in the reverse order, and then subtracted from the original number, the result is 792. Use a system of equations to find all of the three-digit numbers with these properties.

6.2 Exercises: More vectors and matrices

Exercise 6.9 (Compute the action of a matrix on vectors)**.** Let A be the $n \times n$ matrix defined by $A_{i,i+1} = i^2$ for $1 \leq i < n$, and zero otherwise. Compute the action of A and A^T on the standard basis for $\mathbb{R}^n$. Compute the action of A and A^T on an arbitrary vector $\mathbf{x} \in \mathbb{R}^n$.

Exercise 6.10 (Matrix-matrix products, rows and columns)**.** Let $A \in M(n, m, \mathbb{F})$, $B \in M(m, o, \mathbb{F})$. Write B in terms of column vectors, $B = [\mathbf{b}_1, \cdots, \mathbf{b}_o]$. Prove that

$$AB = [A\mathbf{b}_1, \cdots, A\mathbf{b}_o],$$

i.e., A acts on each individual column of B. Similarly, show that, if we write A in terms of its *row vectors*,

$$A = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \cdots \\ \mathbf{a}_n^T \end{bmatrix},$$

then

$$AB = \begin{bmatrix} \mathbf{a}_1^T B \\ \mathbf{a}_2^T B \\ \cdots \\ \mathbf{a}_n^T B \end{bmatrix}.$$

Exercise 6.11. Show that each column of $C = AB$ is a linear combination of the columns of A with coefficients determined by B,

$$\mathbf{c}_i = \sum_j \mathbf{a}_j B_{ji}.$$

Show a similar result for the rows of C.

6.3 Solutions

Solution for Exercise 6.1

a. Vector addition is performed component by component:

$$x + y = (1, 2, -1) + (3, -1, 2) = (4, 1, 1).$$

b. Similarly,

$$x - y = (1, 2, -1) - (3, -1, 2) = (-2, 3, -3).$$

c. Scalar multiplication multiplies every component by the scalar:

$$2x = (2, 4, -2),$$

and

$$-3y = (-9, 3, -6).$$

d. We have

$$2x + 3y = (2, 4, -2) + (9, -3, 6) = (11, 1, 4).$$

e. First,

$$2y = (6, -2, 4),$$

so

$$x + 2y - z = (1, 2, -1) + (6, -2, 4) - (0, 2, 1) = (7, -2, 2).$$

f. We want

$$\alpha x + \beta y = (7, 0, 3).$$

Substituting the vectors gives

$$\alpha(1, 2, -1) + \beta(3, -1, 2) = (7, 0, 3).$$

Comparing components,

$$\alpha + 3\beta = 7, \qquad 2\alpha - \beta = 0, \qquad -\alpha + 2\beta = 3.$$

From the second equation, $\beta = 2\alpha$. Substituting into the first equation gives

$$\alpha + 6\alpha = 7,$$

so $\alpha = 1$ and $\beta = 2$. Indeed,

$$x + 2y = (1, 2, -1) + (6, -2, 4) = (7, 0, 3).$$

g. The left-hand side is

$$2(x + y) = 2(4, 1, 1) = (8, 2, 2).$$

The right-hand side is

$$2x + 2y = (2, 4, -2) + (6, -2, 4) = (8, 2, 2).$$

Thus,

$$2(x+y) = 2x + 2y.$$

This illustrates the distributive law

$$\alpha(x+y) = \alpha x + \alpha y.$$

h. The vector $(1,2)$ is not an element of $\mathbb{R}^3$, since vectors in $\mathbb{R}^3$ have three components.

The vector

$$(0,0,0)$$

is an element of $\mathbb{R}^3$. It is the zero vector of the vector space.

Solution for Exercise 6.2

a. We multiply each row of A by the column vector x:

$$Ax = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \cdot 3 - 1 \cdot 2 \\ 3 \cdot 3 + 4 \cdot 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 17 \end{pmatrix}.$$

b. Similarly,

$$By = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 0 \cdot (-2) + 2 \cdot 3 \\ -1 \cdot 1 + 3 \cdot (-2) + 1 \cdot 3 \end{pmatrix} = \begin{pmatrix} 7 \\ -4 \end{pmatrix}.$$

c. Since

$$2x = \begin{pmatrix} 6 \\ 4 \end{pmatrix},$$

we obtain

$$A(2x) = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 6 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ 34 \end{pmatrix}.$$

From part a,

$$2(Ax) = 2 \begin{pmatrix} 4 \\ 17 \end{pmatrix} = \begin{pmatrix} 8 \\ 34 \end{pmatrix}.$$

Thus $A(2x) = 2(Ax)$.

d. The product Ay cannot be formed because A has two columns while y has three components. The product Bx cannot be formed because B has three columns while x has two components. In general, if A is an $m \times n$ matrix, then Ax is defined when x has n components, and the resulting vector has m components.

Solution for Exercise 6.3

a. We obtain

$$AB = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ -1 & 4 \end{pmatrix} = \begin{pmatrix} 1 \cdot 2 + 2 \cdot (-1) & 1 \cdot 0 + 2 \cdot 4 \\ 3 \cdot 2 + (-1) \cdot (-1) & 3 \cdot 0 + (-1) \cdot 4 \end{pmatrix} = \begin{pmatrix} 0 & 8 \\ 7 & -4 \end{pmatrix}.$$

b. Reversing the order,

$$BA = \begin{pmatrix} 2 & 0 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 11 & -6 \end{pmatrix}.$$

Therefore,

$$AB \neq BA.$$

Matrix multiplication is in general **not commutative**.

c. Since A is 2×2 and C is 2×3, the product AC is a 2×3 matrix:

$$AC = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 2 \end{pmatrix} = \begin{pmatrix} -1 & 8 & 4 \\ 4 & 3 & -2 \end{pmatrix}.$$

d. The product CA cannot be formed: C is 2×3 and A is 2×2, so the inner dimensions, 3 and 2, do not agree. The product BC can be formed because B is 2×2 and C is 2×3. The result would be a 2×3 matrix. In general, the product of an $m \times n$ matrix and an $n \times p$ matrix is defined and gives an $m \times p$ matrix.

Solution to Exercise 6.4

a. Suppose we write $\mathbf{b}_1 x_1 + \mathbf{b}_2 x_2 = 0$. We must show that $x_1 = x_2 = 0$ is the only solution. By adding the equations, we get $3x1 = 0$, and then by insertion into the first equation $x_2 = 0$.
b. Since we have 2 linearly independent vectors in $\mathbb{R}^2$, these form a basis.
c. See further down.

Solution to Exercise 6.5

The equation gives a linear system

$$\begin{bmatrix} b_{11} & b_{21} \\ b_{12} & b_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

where $b_{ij} = (\mathbf{b}_i)_j$. We get an augmented matrix

$$\begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 2 \end{bmatrix}$$

Multiply row 2 by $-1/2$

$$\begin{bmatrix} 1 & -1 & 2 \\ -1 & -1/2 & -1 \end{bmatrix}$$

Add row 1 to row 2:

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & -3/2 & 1 \end{bmatrix}$$

Divide row 2 by $-3/2$:

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -2/3 \end{bmatrix}$$

Add row 2 to row 1:

$$\begin{bmatrix} 1 & 0 & 4/3 \\ 0 & 1 & -2/3 \end{bmatrix}$$

This system is equivalent to

$$x_1 = 4/3, \quad x_2 = -2/3,$$

which is the solution.

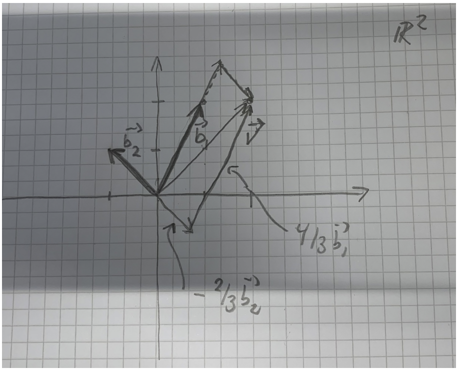


Figure 6.1: image

Solution for Exercise 6.6

Take east as the positive x-direction and north as the positive y-direction. The three displacements are

$$\begin{pmatrix} 0.5 \\ 1 \end{pmatrix} \text{ km}, \qquad \begin{pmatrix} 3 \\ -2 \end{pmatrix} \text{ km}, \qquad \begin{pmatrix} -0.4 \\ -0.2 \end{pmatrix} \text{ km}.$$

The position of the office relative to home is therefore

$$\begin{pmatrix} 0.5 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix} + \begin{pmatrix} -0.4 \\ -0.2 \end{pmatrix} = \begin{pmatrix} 3.1 \\ -1.2 \end{pmatrix} \text{ km}.$$

Thus, the office is 3.1 km east and 1.2 km south of the person's home.

The straight-line distance from home is the Euclidean norm,

$$\sqrt{3.1^2 + (-1.2)^2} = \sqrt{11.05} \text{ km} \approx 3.32 \text{ km}.$$

Solution for Exercise 6.7

Take east as the positive x-direction and north as the positive y-direction. Since the boat travels northeast at 20 km/h relative to the water, its velocity relative to the water is

$$\mathbf{v} = \frac{20}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} = 10\sqrt{2}\begin{pmatrix}1\\1\end{pmatrix} \text{ km/h}.$$

Let $\mathbf{u}$ be the velocity of the ocean drift. After two hours, the observed displacement of the boat is

$$\begin{pmatrix}10\\20\end{pmatrix} \text{ km},$$

so its velocity relative to the ground is

$$\frac{1}{2}\begin{pmatrix}10\\20\end{pmatrix} = \begin{pmatrix}5\\10\end{pmatrix} \text{ km/h}.$$

Therefore,

$$\mathbf{v} + \mathbf{u} = \begin{pmatrix}5\\10\end{pmatrix},$$

and hence

$$\mathbf{u} = \begin{pmatrix}5\\10\end{pmatrix} - 10\sqrt{2}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}5 - 10\sqrt{2}\\10 - 10\sqrt{2}\end{pmatrix} \approx \begin{pmatrix}-9.14\\-4.14\end{pmatrix} \text{ km/h}.$$

The speed of the ocean drift is

$$\|\mathbf{u}\| = \sqrt{(5 - 10\sqrt{2})^2 + (10 - 10\sqrt{2})^2} \approx 10.04 \text{ km/h}.$$

Both components are negative, so the drift is toward the southwest. Its direction is approximately 24.4° south of west.

Solution to Exercise 6.8

Let a be the hundreds digit, b the tens digit, and c the ones digit. The first condition gives

$$b + c = 5.$$

The original number is

$$100a + 10b + c,$$

while the reversed number is

$$100c + 10b + a.$$

The second condition therefore gives

$$(100a + 10b + c) - (100c + 10b + a) = 792,$$

or

$$99a - 99c = 792.$$

Dividing by 99 gives

$$a - c = 8.$$

We therefore have the system

$$a - c = 8, \qquad b + c = 5.$$

As a system over the real numbers, these two equations have infinitely many solutions. However, a, b, and c must be decimal digits, with $a \neq 0$. From

$$a = c + 8,$$

we see that c can only be 0 or 1, since otherwise $a > 9$.

If $c = 0$, then

$$a = 8, \qquad b = 5,$$

giving the number 850.

If $c = 1$, then

$$a = 9, \qquad b = 4,$$

giving the number 941.

Thus, the two possible numbers are

$$\boxed{850 \text{ and } 941}.$$

Solution for Exercise 6.9

The matrix A has the form

$$A = \begin{pmatrix} 0 & 1^2 & 0 & \cdots & 0 \\ 0 & 0 & 2^2 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & (n-1)^2 \\ 0 & \cdots & \cdots & 0 & 0 \end{pmatrix}.$$

Let $\mathbf{e}_1, \dots, \mathbf{e}_n$ be the standard basis vectors of $\mathbb{R}^n$. Since the jth column of A is $A\mathbf{e}_j$, we obtain

$$A\mathbf{e}_1 = 0,$$

and for $2 \leq j \leq n$,

$$A\mathbf{e}_j = (j-1)^2 \mathbf{e}_{j-1}.$$

Similarly, since A^T has entries $(A^T)_{i+1,i} = i^2$,

$$A^T\mathbf{e}_n = 0,$$

and for $1 \leq j < n$,

$$A^T\mathbf{e}_j = j^2\mathbf{e}_{j+1}.$$

Now let

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Using

$$\mathbf{x} = \sum_{j=1}^{n} x_j\mathbf{e}_j,$$

we find

$$A\mathbf{x} = \sum_{j=2}^{n} x_j(j-1)^2\mathbf{e}_{j-1} = \begin{pmatrix} 1^2x_2 \\ 2^2x_3 \\ \vdots \\ (n-1)^2x_n \\ 0 \end{pmatrix}.$$

Thus,

$$(A\mathbf{x})_i = i^2x_{i+1}, \qquad 1 \leq i < n,$$

and

$$(A\mathbf{x})_n = 0.$$

Likewise,

$$A^T\mathbf{x} = \sum_{j=1}^{n-1} x_j j^2 \mathbf{e}_{j+1} = \begin{pmatrix} 0 \\ 1^2 x_1 \\ 2^2 x_2 \\ \vdots \\ (n-1)^2 x_{n-1} \end{pmatrix}.$$

Therefore,

$$(A^T\mathbf{x})_1 = 0,$$

and

$$(A^T\mathbf{x})_i = (i-1)^2 x_{i-1}, \qquad 2 \le i \le n.$$

Solution for Exercise 6.10

Write the columns of B as

$$B = [\mathbf{b}_1, \dots, \mathbf{b}_o].$$

The jth column of the product AB is obtained by multiplying A by the jth column of B. Indeed, its ith component is

$$(AB)_{ij} = \sum_{k=1}^{m} A_{ik} B_{kj}.$$

But the jth column of B is

$$\mathbf{b}_j = \begin{pmatrix} B_{1j} \\ \vdots \\ B_{mj} \end{pmatrix},$$

so

$$(A\mathbf{b}_j)_i = \sum_{k=1}^{m} A_{ik} B_{kj} = (AB)_{ij}.$$

Hence the jth column of AB is $A\mathbf{b}_j$, and therefore

$$AB = [A\mathbf{b}_1, \dots, A\mathbf{b}_o].$$

Now write A in terms of its row vectors,

$$A = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_n^T \end{bmatrix}.$$

The ith row of AB is obtained by multiplying the ith row of A by B. Its jth entry is

$$(AB)_{ij} = \sum_{k=1}^{m} A_{ik} B_{kj}.$$

Since

$$\mathbf{a}_i^T = \begin{pmatrix} A_{i1} & \cdots & A_{im} \end{pmatrix},$$

we have

$$(\mathbf{a}_i^T B)_j = \sum_{k=1}^{m} A_{ik} B_{kj} = (AB)_{ij}.$$

Thus the ith row of AB is $\mathbf{a}_i^T B$, and therefore

$$AB = \begin{bmatrix} \mathbf{a}_1^T B \\ \mathbf{a}_2^T B \\ \vdots \\ \mathbf{a}_n^T B \end{bmatrix}.$$

Solution for Exercise 6.11

Let the columns of A be

$$A = [\mathbf{a}_1, \dots, \mathbf{a}_m],$$

and let the ith column of B be $\mathbf{b}_i$. From the previous exercise, the ith column of $C = AB$ is

$$\mathbf{c}_i = A\mathbf{b}_i.$$

Now

$$\mathbf{b}_i = \begin{pmatrix} B_{1i} \\ B_{2i} \\ \vdots \\ B_{mi} \end{pmatrix},$$

so matrix-vector multiplication gives

$$A\mathbf{b}_i = B_{1i}\mathbf{a}_1 + B_{2i}\mathbf{a}_2 + \cdots + B_{mi}\mathbf{a}_m.$$

Hence

$$\boxed{\mathbf{c}_i = \sum_{j=1}^{m} \mathbf{a}_j B_{ji}}.$$

Thus, each column of C is a linear combination of the columns of A, with coefficients taken from the corresponding column of B.

A similar statement holds for the rows. Let the rows of B be

$$\mathbf{b}_1^T, \dots, \mathbf{b}_m^T,$$

and let the ith row of A be $\mathbf{a}_i^T$. The ith row of $C = AB$ is

$$\mathbf{c}_i^T = \mathbf{a}_i^T B.$$

Since

$$\mathbf{a}_i^T = \begin{pmatrix} A_{i1} & A_{i2} & \cdots & A_{im} \end{pmatrix},$$

we obtain

$$\mathbf{c}_i^T = A_{i1}\mathbf{b}_1^T + A_{i2}\mathbf{b}_2^T + \cdots + A_{im}\mathbf{b}_m^T.$$

Therefore,

$$\mathbf{c}_i^T = \sum_{j=1}^{m} A_{ij}\mathbf{b}_j^T.$$

Thus, each row of C is a linear combination of the rows of B, with coefficients taken from the corresponding row of A.

7 General vector spaces

A vector space needs vector addition and scalar multiplication, but it need not resemble Euclidean coordinate space.

You may be very familiar with Euclidean space $\mathbb{R}^n$ as a vector space, and quite used to the notion that we have an inner product and a corresponding distance measure on this space. However, linear spaces are much more general. An inner product or norm is not part of the definition of a general vector space. Only the a "linear structure" matters: Consider for example a general vector space over $\mathbb{R}$: An abstract set V with operations such that one can *add vectors*,

$$z = x + y \in V,$$

and *multiply vectors by constants* $c \in \mathbb{R}$,

$$z = cx \in V$$

where $c \in \mathbb{R}$. The full set of axioms is as follows. We focus on vector spaces over the field $\mathbb{R}$, but the definition can be generalized to vector spaces over any field such as $\mathbb{C}$.

> **Definition: Vector space.** A *vector space over the field* $\mathbb{R}$ is a set V together with a binary *vector addition* $+ : V \times V \to V$ and *scalar multiplication* $\cdot : \mathbb{R} \times V \to V$ such that, for all $x, y, z \in V$ and all $\alpha, \beta \in \mathbb{R}$, the following axioms are true:
>
> 1. There exists a $0 \in V$ such that $0 + x = x$ for all $x \in V$
> 2. $x + (y + z) = (x + y) + z$
> 3. $x + y = y + x$
> 4. For every $x \in V$, there exists $x' \in V$ such that $x + x' = 0$
> 5. $(\alpha\beta) \cdot x = \alpha \cdot (\beta \cdot x)$
> 6. $1 \cdot x = x$
> 7. $(\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x$

8. $\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y$

In this exercise, we will train our ability to think abstractly, and also try to get acquainted with general vector spaces.

Example: Somerset Apple Cake. Ingredients:

- 170 g butter
- 170 g light brown sugar
- 3 eggs
- 1 tablespoon liquid honey
- 1.5 teaspoons cinnamon
- 0.5 teaspoon cloves
- 0.5 teaspoon nutmeg
- 1 teaspoon baking powder
- 240 g wheat flour
- 700 g apples, diced
- 100 ml apple cider, apple juice, or milk
- 100 g light sultana raisins (optional)

Instructions:

- Cream the room-temperature butter with the sugar until light and fluffy.
- Beat in the eggs and honey.
- Sift the flour, baking powder, and spices. Add them to the mixture and stir until the batter is smooth and lump-free.
- Stir in the milk or apple cider (and the raisins, if you choose to use them; they can be soaked in cider for a couple of hours beforehand, if desired).
- Finally, fold the apple pieces into the batter.
- Pour the batter into a greased round tin with parchment paper at the bottom, or use a deep, round ovenproof dish (24 cm in diameter).

- Bake the cake in the middle of the oven at 160°C for 1.5 hours (check with a cake tester to ensure it's fully baked).
- Serve warm or cold with a dollop of whipped cream, clotted cream, vanilla ice cream, or whatever you like.

7.1 Exercises

Exercise 7.1 (Cooking with vectors 1)**.** We can think of an ingredient list as a vector. For example, suppose that our universe of ingredients consists only of

$$(\text{butter}, \text{sugar}, \text{eggs}, \text{flour}, \text{apples}).$$

After choosing suitable units for each ingredient, an ingredient list can then be represented by a vector

$$x = (x_1, x_2, x_3, x_4, x_5),$$

where each component gives the amount of one ingredient.

a. Using grams for butter, sugar, flour, and apples, and number of eggs for eggs, write the corresponding five-component vector for the Somerset Apple Cake recipe.

b. What would the vector $(100, 0, 2, 300, 0)$ mean? What does a zero component mean?

c. What is the dimension of this vector space? Give a natural set of basis vectors and explain what each basis vector represents.

d. Suppose that x and y are the ingredient vectors for two recipes. What does the vector $x + y$ represent?

e. What does the vector $2x$ represent? What about $\frac{1}{2}x$? Does the scalar 2 have a physical unit?

f. Our choice of units is part of the way we represent the recipe. Suppose milk were another component. Explain why we should not represent 1 litre of milk as 1 in one recipe and 1000 ml of milk as 1000 in another without taking account of the different units.

g. Typical recipes contain ingredients that are not among our five chosen ingredients. How could we enlarge our vector space to accommodate more recipes? What happens to the dimension of the space?

Exercise 7.2 (Cooking with vectors 2)**.** In Exercise 7.1, we represented recipes using a fixed list of possible ingredients. We can make this construction more general.

Let S be the set of all possible ingredients. An ingredient list can be represented by a function

$$f : S \to \mathbb{R},$$

where $f(s)$ gives the amount of ingredient s. We assume that $f(s) = 0$ for all but finitely many ingredients. The set L of all such functions is a vector space, with addition and scalar multiplication defined in the usual way.

a. Explain how this description generalizes the finite-dimensional ingredient vectors used in Exercise 7.1. If S contains infinitely many ingredients, what can you say about the dimension of L?

b. Mathematically, L contains vectors with negative components. Can such vectors be interpreted as ordinary ingredient lists? What about the zero vector?

c. Define the subset

$$L_+ = \{f \in L \mid f(s) \geq 0 \text{ for every } s \in S\}.$$

Is L_+ a vector space? Which vector-space properties fail? Is L_+ closed under addition? Is it closed under multiplication by *nonnegative* scalars?

d. Even if $f, g \in L_+$ are ingredient lists for two perfectly good recipes, $f + g$ need not describe a sensible recipe. What information is missing from an ingredient vector?

e. We might try to describe the steps of a recipe by operators acting on L. Consider, for example, the instruction "cream the butter and sugar". Can this operation be represented naturally as a linear map $T : L \to L$ on our present ingredient space? Explain the challenges one encounters.

f. What additional information might have to be included in our mathematical description before operations such as mixing, creaming, heating, and baking could be represented?

g. In quantum mechanics, we similarly choose a mathematical space to represent physical states and operators to represent physical quantities and transformations. Why is the choice of the space important for determining what the mathematical model can describe?

7.2 Solutions

Solution for Exercise 7.1

a. With the ordering

$$(\text{butter}, \text{sugar}, \text{eggs}, \text{flour}, \text{apples}),$$

the Somerset Apple Cake recipe is represented by

$$x = (170, 170, 3, 240, 700).$$

The components do not all have the same units: the first, second, fourth, and fifth components are measured in grams, while the third is a number of eggs. This is not a problem as long as the meaning and unit of each component are fixed.

b. The vector

$$(100, 0, 2, 300, 0)$$

represents an ingredient list containing 100 g butter, no sugar, 2 eggs, 300 g flour, and no apples.

A zero component means that the corresponding ingredient is absent.

c. The vector space has dimension 5. A natural basis is

$$e_1 = (1, 0, 0, 0, 0), \qquad e_2 = (0, 1, 0, 0, 0), \qquad \dots, \qquad e_5 = (0, 0, 0, 0, 1).$$

Each basis vector represents one unit of one ingredient and zero of all the others. For example, e_1 represents 1 g of butter and e_3 represents 1 egg.

Any ingredient vector can be written as a linear combination of these basis vectors. For example,

$$(170, 170, 3, 240, 700) = 170e_1 + 170e_2 + 3e_3 + 240e_4 + 700e_5.$$

d. Addition is performed component by component. If

$$x = (x_1, \dots, x_5), \qquad y = (y_1, \dots, y_5),$$

then

$$x + y = (x_1 + y_1, \dots, x_5 + y_5).$$

Thus, $x + y$ represents putting together the ingredient quantities from the two lists. If both recipes require flour, for example, the flour amounts are added.

The resulting vector is certainly a valid ingredient vector, although it need not correspond to a sensible cake recipe.

e. Multiplying by 2 doubles the amount of every ingredient:

$$2x = (2x_1, \dots, 2x_5).$$

This can be interpreted as making twice as much of the recipe. Similarly,

$$\frac{1}{2}x$$

corresponds to making half as much.

The scaling factor is dimensionless. It simply tells us by what numerical factor all ingredient quantities are changed.

f. The numerical value of a component only has meaning together with the unit chosen for that component. If the milk coordinate is defined to be measured in millilitres, then

$$1 \text{ litre} = 1000 \text{ ml}$$

must be represented by the number 1000.

If we sometimes used litres and sometimes millilitres without converting them, then the same physical amount could be represented by different vectors. We therefore choose one standard unit for each component.

g. We can add more components, one for each new ingredient. For example, if cinnamon and milk are added to our list of possible ingredients, the vectors would have seven components instead of five.

Each additional independent ingredient increases the dimension by one. If we allow n possible ingredients, the corresponding vector space has dimension n.

Solution for Exercise 7.2

a. In the previous exercise, we first chose a finite ordered list of ingredients and represented an ingredient list by a vector in $\mathbb{R}^n$.

The function description achieves the same without requiring us to choose a finite list in advance. For every possible ingredient $s \in S$, the number $f(s)$ specifies how much of that ingredient is present.

If S is infinite, the space L is infinite dimensional. A natural basis consists of functions e_s, one for every ingredient $s \in S$, defined by

$$e_s(t) = \begin{cases} 1, & t = s, \\ 0, & t \neq s. \end{cases}$$

An ingredient list with finitely many ingredients can then be written as a finite linear combination of these basis functions.

b. Mathematically, negative values are perfectly allowed in the vector space L. For example, a vector with

$$f(\text{flour}) = -200$$

is a valid vector in L.

It is not, however, an ordinary ingredient list, since a recipe cannot contain negative amounts of flour.

The zero vector, for which

$$f(s) = 0$$

for every ingredient s, is mathematically valid and can be interpreted as a very healthy cake - an empty ingredient list.

c. The set

$$L_+ = \{f \in L \mid f(s) \geq 0 \text{ for every } s \in S\}$$

is not a vector space. It is, however, the natural set of ingredient lists, since a recipe cannot contain negative amounts of any ingredient.

It is closed under addition: if $f(s) \geq 0$ and $g(s) \geq 0$, then

$$(f + g)(s) = f(s) + g(s) \geq 0.$$

It is also closed under multiplication by a nonnegative scalar $c \geq 0$, since

$$(cf)(s) = cf(s) \geq 0.$$

However, it is not closed under multiplication by negative scalars. If f is a nonzero ingredient list, then $-f$ has negative components and therefore does not belong to L_+.

Equivalently, nonzero elements of L_+ do not have additive inverses within L_+. Thus L_+ fails one of the essential requirements for being a vector space.

d. An ingredient vector specifies only which ingredients occur and how much of each one is present. It does not specify, for example,

   - the order in which ingredients are combined,
   - whether ingredients are mixed, folded, beaten, or creamed,
   - temperatures and cooking times,
   - the physical state of the ingredients,
   - whether a particular combination of ingredients is actually edible or sensible.

   Consequently, $f + g$ is always a mathematically well-defined vector, but it need not correspond to a meaningful recipe.

   The full set L is guaranteed to be a vector space because it was constructed to satisfy the vector-space axioms. The subset L_+ is not a vector space, and the still smaller set of sensible recipes is certainly not guaranteed to be one either.

e. Not naturally, at least not on the space L as presently defined.

   The vector $f \in L$ records only amounts of ingredients. After butter and sugar have been creamed together, their physical state has changed, but our vector space has no component that records whether the butter and sugar are separate or have been creamed together.

   One could artificially introduce a new component such as

   "creamed butter and sugar",

   but this already shows that the original state space was insufficient. It also raises the question of just how "universal" the set S should be. Should it contain "creamed butter and sugar" as a separate ingredient, and should it contain objects like "50 % butter and 54 % sugar creamed for 10 minutes at 302 K and 1 atm"?

   There is also no obvious reason why the actual physical process of creaming should satisfy linearity,

$$T(\alpha f + \beta g) = \alpha T(f) + \beta T(g).$$

   Thus the main problem is not merely finding a matrix for the operation: the space L does not contain enough information to describe the process.

f. We would need a richer description of the state. Depending on how detailed the model is, this might include quantities such as

   - which ingredients have already been combined,
   - temperature,
   - physical phase or consistency,
   - degree of mixing,
   - particle or droplet size,
   - chemical composition,
   - elapsed cooking time.

   The appropriate mathematical state space depends on which properties and transformations we want the model to describe.

g. The same general idea appears in quantum mechanics. Before an operator can act on a state, we must first specify what mathematical objects represent the possible states.

   In quantum mechanics, states are represented by vectors in a suitable vector space, some kind of Hilbert space, and observables are represented by linear operators on that space.

   The choice of state space determines which states can be represented and which operators can act on them. If some physically relevant information is absent from the state description, an operator acting only on that space cannot describe transformations involving that missing information.

   Thus, both in the recipe example and in quantum mechanics, choosing the vector space is not merely a matter of notation - it is part of constructing the mathematical model of reality.

8 The complex numbers as a real vector space

Identifying $x + \mathrm{i}y$ with (x, y) turns the complex plane into a two-dimensional real vector space.

The geometrical interpretation of the $\mathbb{C}$ as the plane $\mathbb{R}^2$ indicates that $\mathbb{C}$ can be viewed as a two-dimensional real vector space. Indeed, addition and multiplication with *real* scalars are compatible with the axioms for Euclidean space $\mathbb{R}^2$.

8.1 Exercises

Exercise 8.1 (Complex numbers as a real vector space).

a. Show that $\mathbb{C}$ regarded can be regarded as the Euclidean plane: Check axioms for Euclidean space and check that vector addition and multiplication with real scalars are the same in the two spaces. Check also that the Euclidean norm is the modulus of the complex number. What are the complex numbers that correspond to the standard basis in $\mathbb{R}^2$? Conclude that $\mathbb{C}$ can be regarded as a *real* Euclidean vector space of dimension 2.

b. Show that multiplication with a *complex* number z is a linear operator on $\mathbb{R}^2$, and find its matrix in the standard basis. What is the matrix of multiplication with i?

c. Show that multiplication with a complex number of modulus 1 is a rotation in $\mathbb{R}^2$.

d. Show that the map $z \mapsto \bar{z}$ is a linear transformation in $\mathbb{R}^2$. What kind of linear transformation is this? Is it a linear transformation in $\mathbb{C}$?

8.2 Solutions

Solution for Exercise 8.1

a. Write a complex number as

$$z = x + \mathrm{i}y, \qquad x, y \in \mathbb{R}.$$

Associate z with the vector

$$\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2.$$

If

$$z_1 = x_1 + \mathrm{i}y_1, \qquad z_2 = x_2 + \mathrm{i}y_2,$$

then

$$z_1 + z_2 = (x_1 + x_2) + \mathrm{i}(y_1 + y_2),$$

which corresponds exactly to

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}.$$

Similarly, for a real scalar α,

$$\alpha z = \alpha x + \mathrm{i}\alpha y,$$

corresponding to

$$\alpha \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha x \\ \alpha y \end{pmatrix}.$$

Thus vector addition and multiplication by real scalars agree with the usual operations in $\mathbb{R}^2$. Since these operations are exactly those of $\mathbb{R}^2$, the real vector-space axioms are satisfied.

The Euclidean norm of the corresponding vector is

$$\left\|\begin{pmatrix} x \\ y \end{pmatrix}\right\| = \sqrt{x^2 + y^2},$$

which is exactly the modulus of the complex number,

$$|z| = \sqrt{x^2 + y^2}.$$

The standard basis vectors in $\mathbb{R}^2$,

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

correspond to the complex numbers

$$1, \qquad \mathrm{i}.$$

Therefore,

$$\mathbb{C} = \operatorname{span}_{\mathbb{R}}\{1, \mathrm{i}\},$$

and $\mathbb{C}$ can be regarded as a real Euclidean vector space of dimension 2.

b. Let

$$z = a + \mathrm{i}b$$

be fixed, and define

$$T_z(w) = zw.$$

For real scalars α, β and complex numbers w_1, w_2,

$$T_z(\alpha w_1 + \beta w_2) = z(\alpha w_1 + \beta w_2) = \alpha z w_1 + \beta z w_2 = \alpha T_z(w_1) + \beta T_z(w_2).$$

Hence multiplication by a fixed complex number is linear when $\mathbb{C}$ is regarded as a vector space over $\mathbb{R}$.

Now let

$$w = x + \mathrm{i}y.$$

Then

$$zw = (a + \mathrm{i}b)(x + \mathrm{i}y) = (ax - by) + \mathrm{i}(bx + ay).$$

Therefore,

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} ax - by \\ bx + ay \end{pmatrix},$$

so the matrix of multiplication by $z = a + \mathrm{i}b$ in the standard basis is

$$[T_z] = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

For $z = \mathrm{i}$, we have $a = 0$ and $b = 1$, giving

$$[T_\mathrm{i}] = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

c. Let z have modulus 1. Then it can be written as

$$z = \cos\theta + \mathrm{i}\sin\theta.$$

The matrix of multiplication by z is therefore

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

This is exactly the standard rotation matrix through the angle θ. Hence multiplication by a complex number of modulus 1 corresponds to a rotation in $\mathbb{R}^2$.

d. Define complex conjugation by

$$T(z) = \bar{z}.$$

If

$$z = x + \mathrm{i}y,$$

then

$$\bar{z} = x - \mathrm{i}y.$$

Thus, in $\mathbb{R}^2$,

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ -y \end{pmatrix}.$$

Its matrix is therefore

$$[T] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For real scalars α, β,

$$\overline{\alpha z_1 + \beta z_2} = \alpha \bar{z}_1 + \beta \bar{z}_2,$$

so complex conjugation is linear over $\mathbb{R}$.

Geometrically, it is a reflection in the real axis.

It is not linear over $\mathbb{C}$. For example, complex linearity would require

$$T(\mathrm{i}z) = \mathrm{i}T(z),$$

but

$$T(\mathrm{i}z) = \overline{\mathrm{i}z} = -\mathrm{i}\bar{z},$$

whereas

$$\mathrm{i}T(z) = \mathrm{i}\bar{z}.$$

These are not equal in general. Thus complex conjugation is real-linear but not complex-linear.

9 Gaussian elimination

9.1 Exercises

Exercise 9.1 (Gaussian elimination and overlap matrix)**.** Consider the two vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

a. Compute the overlap matrix S with elements

$$S_{ij} = \mathbf{v}_i^T \mathbf{v}_j.$$

b. Verify that S is symmetric.

c. Use Gaussian elimination on the augmented matrix $[S \,|\, I]$ to compute S^{-1}.

d. Verify your result by computing SS^{-1}.

Exercise 9.2. Write down the dual basis of $B = \{\mathbf{b}_1, \mathbf{b}_2\}$.

Exercise 9.3. (Robert Beezer) Consider the following system of equations:

$$\begin{aligned} 2x_1 - 3x_2 + x_3 + 7x_4 &= 14 \\ 2x_1 + 8x_2 - 4x_3 + 5x_4 &= -1 \\ x_1 + 3x_2 - 3x_3 &= 4 \\ -5x_1 + 2x_2 + 3x_3 + 4x_4 &= -19 \end{aligned}$$

Use Gaussian elimination to find all possible solutions. Write the solution set using set notation.

Exercise 9.4. (Robert Beezer) Find all possible solutions of the linear system

$$\begin{aligned}3x_1 + 4x_2 - x_3 + 2x_4 &= 6\\ x_1 - 2x_2 + 3x_3 + x_4 &= 2\\ 10x_2 - 10x_3 - x_4 &= 1\end{aligned}$$

Write down the solution set using set notation.

Exercise 9.5. (Robert Beezer) Find all possible solutions of the linear system

$$\begin{aligned}2x_1 + 4x_2 + 5x_3 + 7x_4 &= -26\\ x_1 + 2x_2 + x_3 - x_4 &= -4\\ -2x_1 - 4x_2 + x_3 + 11x_4 &= -10\end{aligned}$$

Write down the solution set using set notation.

Exercise 9.6. Let $\mathbf{b}_i$, $i = 1, 2, \cdots, n$ be a basis for $\mathbb{R}^n$. Let

$$B = [\mathbf{b}_1, \mathbf{b}_2, \cdots, \mathbf{b}_n]$$

be the *basis matrix*, whose columns are precisely the $\mathbf{b}_i$. Explain that if $\mathbf{y} = x_1\mathbf{b}_1 + \cdots + x_n\mathbf{b}_n$, then

$$B\mathbf{x} = \mathbf{y}.$$

Solve for $\mathbf{x}$ using B in symbols.

Exercise 9.7. Let n linearly independent vectors $B = \{\mathbf{b}_1, \cdots, \mathbf{b}_n\}$ of $\mathbb{F}^n$ be given. Write down a formula for the dual basis in terms of the overlap matrix.

9.2 Solutions

Solution for Exercise 9.1

a. The overlap matrix is defined by

$$S_{ij} = \mathbf{v}_i^T\mathbf{v}_j.$$

We have

$$\mathbf{v}_1^T\mathbf{v}_1 = 2, \qquad \mathbf{v}_1^T\mathbf{v}_2 = 1, \qquad \mathbf{v}_2^T\mathbf{v}_1 = 1, \qquad \mathbf{v}_2^T\mathbf{v}_2 = 2.$$

Therefore,

$$S = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

b. Since

$$S^T = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = S,$$

the overlap matrix is symmetric.

c. To find the inverse, we perform Gaussian elimination on the augmented matrix

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right].$$

Interchange the two rows:

$$\left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array}\right].$$

Now perform $R_2 \leftarrow R_2 - 2R_1$:

$$\left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & -3 & 1 & -2 \end{array}\right].$$

Divide the second row by -3:

$$\left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & 1 & -\frac{1}{3} & \frac{2}{3} \end{array}\right].$$

Finally, perform $R_1 \leftarrow R_1 - 2R_2$:

$$\left[\begin{array}{cc|cc} 1 & 0 & \frac{2}{3} & -\frac{1}{3} \\ 0 & 1 & -\frac{1}{3} & \frac{2}{3} \end{array}\right].$$

Hence,

$$S^{-1} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

d. We verify that

$$SS^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus the inverse is correct.

Solution to Exercise 9.2

The source does not supply a solution.

Solution to Exercise 9.3

Using elementary operations, we get the augmented matrix

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & -4 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right].$$

The solution is unique: $x_1 = 1$, $x_2 = -3$, $x_3 = -4$, and $x_4 = 1$. Thus

$$S = \{[1, -3, -4, 1]^T\}.$$

Solution to Exercise 9.4

By elementary row operations, we obtain the augmented matrix:

$$\begin{bmatrix} 1 & 0 & 1 & 4/5 & 0 \\ 0 & 1 & -1 & -1/10 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

The last row represents the equation $0 = 1$, so there cannot be a solution to the system. $S = \{\} = \emptyset$.

Solution to Exercise 9.5

The augmented matrix reduces to

$$\begin{pmatrix} 1 & 2 & 0 & -4 & 2 \\ 0 & 0 & 1 & 3 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Hence x_2 and x_4 are free, while

$$x_1 = -2x_2 + 4x_4 + 2, \qquad x_3 = -3x_4 - 6.$$

The solution set is

$$S = \{[-2x_2 + 4x_4 + 2, x_2, -3x_4 - 6, x_4]^T \mid x_2, x_4 \in \mathbb{F}\}.$$

Solution to Exercise 9.6

We know that the overlap matrix should be key here. So we multiply with B^T from the left to get $B^T B\mathbf{x} = B^T\mathbf{y}$. Thus,

$$\mathbf{x} = S^{-1}B^T\mathbf{y}.$$

We know that the overlap matrix $S = B^T B$ is invertible, since B is a basis matrix.

Solution to Exercise 9.7

The dual basis $\tilde{B} = [\tilde{\mathbf{b}}_1, \cdots \tilde{\mathbf{b}}_n]$ is such that $\langle \tilde{\mathbf{b}}_i, \mathbf{b}_j \rangle = \delta_{ij}$. We obtain the equation

$$\tilde{B}^H B = I.$$

Thus,

$$\tilde{B} = (B^{-1})^H.$$

10 Bra-ket notation

Bra-ket notation, introduced by Dirac, is a compact way of writing the linear algebra of quantum mechanics. A vector in a Hilbert space is written as a *ket*,

$$|\psi\rangle,$$

and its Hermitian adjoint is a *bra*,

$$\langle\psi|.$$

The inner product of two vectors is therefore written

$$\langle\phi|\psi\rangle,$$

and is a scalar. It satisfies

$$\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*.$$

In particular,

$$\langle\psi|\psi\rangle = \|\psi\|^2,$$

so a normalized state satisfies $\langle\psi|\psi\rangle = 1$.

The product in the opposite order,

$$|\phi\rangle\langle\psi|,$$

is an *outer product*. It is an operator: acting on a ket $|\chi\rangle$ it gives

$$\big(|\phi\rangle\langle\psi|\big)|\chi\rangle = |\phi\rangle\langle\psi|\chi\rangle.$$

A linear operator $\hat{A}$ acts on kets from the left,

$$|\phi\rangle = \hat{A}|\psi\rangle.$$

Taking the Hermitian adjoint reverses the order,

$$\langle\phi| = \langle\psi|\hat{A}^\dagger.$$

For a Hermitian operator, $\hat{A}^\dagger = \hat{A}$. Observables in quantum mechanics are represented by Hermitian operators, and the expectation value of $\hat{A}$ in a normalized state $|\psi\rangle$ is

$$\langle A \rangle = \langle \psi | \hat{A} | \psi \rangle.$$

For an orthonormal basis $\{|i\rangle\}$,

$$\langle i | j \rangle = \delta_{ij},$$

and every vector can be expanded as

$$|\psi\rangle = \sum_i c_i |i\rangle, \qquad c_i = \langle i | \psi \rangle.$$

Equivalently, the basis satisfies the completeness relation

$$\hat{I} = \sum_i |i\rangle\langle i|.$$

In quantum chemistry we also frequently use *nonorthogonal* basis functions, such as atomic orbitals. If the basis is $\{|b_i\rangle\}$, its overlap matrix is

$$S_{ij} = \langle b_i | b_j \rangle.$$

In this case the expansion coefficients cannot in general be obtained simply by projecting with $\langle b_i|$; the overlap matrix must also be taken into account.

10.1 Exercises

Exercise 10.1 (Bra-ket notation and column vectors)**.** Consider the orthonormal basis $\{|1\rangle, |2\rangle, |3\rangle\}$. In this basis, the ket

$$|\psi\rangle = 2|1\rangle - i|2\rangle + 3|3\rangle$$

can be represented by a column vector.

a. Write $|1\rangle$, $|2\rangle$, and $|3\rangle$ as column vectors.

b. Write $|\psi\rangle$ as a column vector.

c. Write the bra $\langle\psi|$ in bra-ket notation.

d. Write $\langle\psi|$ as a row vector. Remember that taking the adjoint involves both transposing and complex conjugating.

e. Calculate $\langle 2|\psi\rangle$, first using bra-ket notation and then by multiplying a row vector by a column vector.

f. Calculate $\langle\psi|\psi\rangle$ using the row and column vectors. What is the norm $\|\psi\|$?

Exercise 10.2 (Reading simple bra-ket expressions)**.** Consider two orthonormal states $|1\rangle$ and $|2\rangle$ and the state

$$|\psi\rangle = c_1|1\rangle + c_2|2\rangle.$$

a. Write the corresponding bra $\langle\psi|$.

b. Compute $\langle 1|\psi\rangle$.

c. Compute $\langle 2|\psi\rangle$.

d. Compute $\langle\psi|\psi\rangle$.

e. What condition must c_1 and c_2 satisfy if $|\psi\rangle$ is normalized?

Exercise 10.3 (From coordinate vectors to Hilbert-space vectors)**.** Let $B = \{|b_1\rangle, \dots, |b_n\rangle\}$ be a basis for an n-dimensional Hilbert space V, and let $\{|i\rangle\}$ be the standard orthonormal basis of $\mathbb{C}^n$. Define

$$\hat{B} = \sum_{i=1}^{n} |b_i\rangle\langle i|.$$

a. What are the domain and range of $\hat{B}$?

b. Let

$$|x\rangle = \sum_i x_i|i\rangle \in \mathbb{C}^n.$$

Show that

$$\hat{B}|x\rangle = \sum_i x_i|b_i\rangle.$$

Interpret the numbers x_i.

c. Find $\hat{B}^\dagger$.

d. Show that

$$\hat{B}^\dagger\hat{B} = \sum_{ij} |i\rangle\langle b_i|b_j\rangle\langle j|.$$

e. Explain why $\hat{B}^\dagger\hat{B}$ is the overlap matrix of the basis B, expressed as an operator on $\mathbb{C}^n$.

Exercise 10.4 (Molecular orbitals in a nonorthogonal atomic-orbital basis). Suppose a molecular orbital is expanded in two normalized atomic orbitals,

$$|\psi\rangle = c_1|\chi_1\rangle + c_2|\chi_2\rangle,$$

where

$$\langle\chi_1|\chi_1\rangle = \langle\chi_2|\chi_2\rangle = 1, \qquad \langle\chi_1|\chi_2\rangle = \langle\chi_2|\chi_1\rangle = s,$$

with s real.

a. Compute $\langle\psi|\psi\rangle$.

b. Write the overlap matrix S in the basis $\{|\chi_1\rangle, |\chi_2\rangle\}$.

c. Show that the normalization condition can be written

$$\mathbf{c}^\dagger S\mathbf{c} = 1,$$

where $\mathbf{c} = (c_1, c_2)^T$.

d. Consider the symmetric combination

$$|\psi_+\rangle = N_+\big(|\chi_1\rangle + |\chi_2\rangle\big).$$

Find the normalization constant N_+.

e. Similarly, find the normalization constant N_- for

$$|\psi_-\rangle = N_-\big(|\chi_1\rangle - |\chi_2\rangle\big).$$

f. Show that $\langle\psi_+|\psi_-\rangle = 0$. Thus symmetric and antisymmetric combinations can be orthogonal even though the atomic orbitals from which they are constructed are not.

Exercise 10.5 (Nonorthogonal bases, dual vectors, and completeness). Let $\{|b_i\rangle\}_{i=1}^n$ be a linearly independent, but not necessarily orthogonal, basis of an n-dimensional Hilbert space. Define the overlap matrix

$$S_{ij} = \langle b_i|b_j\rangle.$$

Assume that S is invertible.

a. Define vectors $|b^i\rangle$ by

$$|b^i\rangle = \sum_j |b_j\rangle (S^{-1})_{ji}.$$

Show that

$$\langle b_k | b^i \rangle = \delta^i_k.$$

The set $\{|b^i\rangle\}$ is called the *dual basis*.

b. Show that

$$\hat{I} = \sum_i |b^i\rangle\langle b_i|$$

and hence also

$$\hat{I} = \sum_{ij} |b_i\rangle (S^{-1})_{ij} \langle b_j|.$$

c. Let

$$|\psi\rangle = \sum_i c_i |b_i\rangle.$$

Show that the expansion coefficients are

$$c_i = \langle b^i | \psi \rangle.$$

d. Define the projections

$$p_i = \langle b_i | \psi \rangle.$$

Show that

$$p_i = \sum_j S_{ij} c_j$$

and therefore, in matrix notation,

$$\mathbf{c} = S^{-1}\mathbf{p}.$$

e. Explain why the familiar orthonormal-basis formula $c_i = \langle b_i | \psi \rangle$ is recovered when $S = I$.

f. In quantum chemistry, atomic-orbital basis functions are generally nonorthogonal. Explain briefly why confusing the projections $\langle b_i | \psi \rangle$ with the expansion coefficients c_i can therefore lead to incorrect molecular-orbital coefficients.

10.2 Solutions

Solution for Exercise 10.1

a. The basis kets are represented by the standard basis vectors:

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad |2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad |3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

b. The coefficients in the expansion of $|\psi\rangle$ become the entries of the column vector:

$$|\psi\rangle = \begin{pmatrix} 2 \\ -i \\ 3 \end{pmatrix}.$$

c. To form the bra, we take the Hermitian adjoint, so the coefficients are complex conjugated:

$$\langle\psi| = 2\langle 1| + i\langle 2| + 3\langle 3|.$$

d. The corresponding row vector is

$$\langle\psi| = \begin{pmatrix} 2 & i & 3 \end{pmatrix}.$$

Thus, in matrix notation, changing a ket into a bra means taking the conjugate transpose:

$$|\psi\rangle^\dagger = \langle\psi|.$$

e. Since the basis is orthonormal,

$$\langle 2|\psi\rangle = 2\langle 2|1\rangle - i\langle 2|2\rangle + 3\langle 2|3\rangle = -i.$$

Using row and column vectors gives the same result:

$$\langle 2|\psi\rangle = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ -i \\ 3 \end{pmatrix} = -i.$$

f. Using the row and column vectors,

$$\langle\psi|\psi\rangle = \begin{pmatrix}2 & i & 3\end{pmatrix}\begin{pmatrix}2\\ -i\\ 3\end{pmatrix} = 4+1+9 = 14.$$

Therefore,

$$\|\psi\| = \sqrt{\langle\psi|\psi\rangle} = \sqrt{14}.$$

Solution for Exercise 10.2

a. Taking the Hermitian adjoint conjugates the coefficients:

$$\langle\psi| = c_1^*\langle 1| + c_2^*\langle 2|.$$

b. Since the basis is orthonormal,

$$\langle 1|\psi\rangle = c_1\langle 1|1\rangle + c_2\langle 1|2\rangle = c_1.$$

c. Similarly,

$$\langle 2|\psi\rangle = c_2.$$

d. We have

$$\langle\psi|\psi\rangle = (c_1^*\langle 1| + c_2^*\langle 2|)(c_1|1\rangle + c_2|2\rangle).$$

Using $\langle i|j\rangle = \delta_{ij}$ gives

$$\langle\psi|\psi\rangle = |c_1|^2 + |c_2|^2.$$

e. For a normalized state,

$$|c_1|^2 + |c_2|^2 = 1.$$

Solution for Exercise 10.3

a. The operator $\hat{B}$ maps coordinate vectors in $\mathbb{C}^n$ to vectors in V. Thus its domain is $\mathbb{C}^n$ and its range is V. Because the vectors $\{|b_i\rangle\}$ form a basis, every vector in V is in the range.

b. Using the definition of $|x\rangle$,

$$\hat{B}|x\rangle = \sum_{ij} |b_i\rangle\langle i|x_j|j\rangle = \sum_{ij} x_j|b_i\rangle\delta_{ij} = \sum_i x_i|b_i\rangle.$$

Thus the components x_i of the coordinate vector are precisely the expansion coefficients of the corresponding vector in the basis B.

c. Taking the Hermitian adjoint gives

$$\hat{B}^\dagger = \sum_i |i\rangle\langle b_i|.$$

d. Therefore

$$\hat{B}^\dagger\hat{B} = \left(\sum_i |i\rangle\langle b_i|\right)\left(\sum_j |b_j\rangle\langle j|\right) = \sum_{ij} |i\rangle\langle b_i|b_j\rangle\langle j|.$$

e. The matrix element of $\hat{B}^\dagger\hat{B}$ in the standard basis is

$$\langle i|\hat{B}^\dagger\hat{B}|j\rangle = \langle b_i|b_j\rangle = S_{ij}.$$

Hence $\hat{B}^\dagger\hat{B}$ is exactly the overlap matrix S represented as an operator on the coordinate space $\mathbb{C}^n$.

Solution for Exercise 10.4

a. We have

$$\langle\psi| = c_1^*\langle\chi_1| + c_2^*\langle\chi_2|.$$

Therefore

$$\langle\psi|\psi\rangle = |c_1|^2 + |c_2|^2 + s(c_1^*c_2 + c_2^*c_1).$$

Equivalently,

$$\langle\psi|\psi\rangle = |c_1|^2 + |c_2|^2 + 2s\ \mathrm{Re}(c_1^*c_2).$$

b. The overlap matrix is

$$S = \begin{pmatrix} 1 & s \\ s & 1 \end{pmatrix}.$$

c. With

$$\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

we obtain

$$\mathbf{c}^\dagger S \mathbf{c} = \begin{pmatrix} c_1^* & c_2^* \end{pmatrix} \begin{pmatrix} 1 & s \\ s & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

which is exactly

$$|c_1|^2 + |c_2|^2 + s(c_1^* c_2 + c_2^* c_1).$$

Thus normalization requires

$$\mathbf{c}^\dagger S \mathbf{c} = 1.$$

d. For the symmetric combination,

$$1 = \langle \psi_+ | \psi_+ \rangle = |N_+|^2 (2 + 2s).$$

Choosing N_+ real and positive gives

$$N_+ = \frac{1}{\sqrt{2(1+s)}}.$$

e. Similarly,

$$1 = \langle \psi_- | \psi_- \rangle = |N_-|^2 (2 - 2s),$$

so

$$N_- = \frac{1}{\sqrt{2(1-s)}}.$$

f. We find

$$\langle \psi_+ | \psi_- \rangle = N_+ N_- \big(\langle \chi_1 | + \langle \chi_2 | \big) \big(| \chi_1 \rangle - | \chi_2 \rangle \big).$$

Hence

$$\langle \psi_+ | \psi_- \rangle = N_+ N_- (1 - s + s - 1) = 0.$$

The symmetric and antisymmetric combinations are therefore orthogonal.

Solution for Exercise 10.5

a. From the definition,

$$\langle b_k | b^i \rangle = \sum_j \langle b_k | b_j \rangle (S^{-1})_{ji} = \sum_j S_{kj} (S^{-1})_{ji} = \delta_k^i.$$

b. Let the proposed operator act on an arbitrary basis vector $|b_k\rangle$:

$$\left(\sum_i |b^i\rangle\langle b_i| \right) |b_k\rangle = \sum_i |b^i\rangle S_{ik}.$$

Substituting the definition of the dual vectors,

$$\sum_{ij} |b_j\rangle (S^{-1})_{ji} S_{ik} = \sum_j |b_j\rangle \delta_{jk} = |b_k\rangle.$$

Since the operator acts as the identity on every basis vector, it is the identity operator:

$$\hat{I} = \sum_i |b^i\rangle\langle b_i|.$$

Substituting

$$|b^i\rangle = \sum_j |b_j\rangle (S^{-1})_{ji}$$

gives

$$\hat{I} = \sum_{ij} |b_i\rangle (S^{-1})_{ij} \langle b_j|.$$

c. Apply $\langle b^i|$ to the expansion:

$$\langle b^i | \psi \rangle = \sum_j c_j \langle b^i | b_j \rangle = \sum_j c_j \delta_j^i = c_i.$$

Thus

$$c_i = \langle b^i | \psi \rangle.$$

d. Projecting $|\psi\rangle$ onto $\langle b_i|$ gives

$$p_i = \langle b_i | \psi \rangle = \sum_j c_j \langle b_i | b_j \rangle = \sum_j S_{ij} c_j.$$

Hence

$$\mathbf{p} = S\mathbf{c},$$

and therefore

$$\mathbf{c} = S^{-1}\mathbf{p}.$$

e. For an orthonormal basis,

$$S_{ij} = \delta_{ij},$$

so $S = I$ and $S^{-1} = I$. The dual basis is then identical to the original basis, and

$$c_i = \langle b_i | \psi \rangle.$$

f. For nonorthogonal atomic orbitals,

$$\langle b_i | \psi \rangle = \sum_j S_{ij} c_j,$$

so a projection onto one basis function contains contributions from several expansion coefficients. The projections equal the coefficients only when the basis is orthonormal. In a nonorthogonal atomic-orbital basis, the overlap matrix must therefore be accounted for when recovering the molecular-orbital coefficients.

11 Eigenvalues and diagonalization

These exercises develop the spectral theorem and apply determinants to finite-dimensional eigenvalue problems.

11.1 Exercises

11.1.1 Eigenvalue decomposition

Of great importance to quantum chemistry is the time-independent Schrödinger equation: Given the Hamiltonian operator $\hat{H}$, which is usually Hermitian, find a nonzero $|\psi\rangle$, called an eigenvector, and a real number E, called an eigenvalue, such that

$$\hat{H}\,|\psi\rangle = E\,|\psi\rangle\,.$$

Choosing a particular orthonormal basis $\{|\phi_i\rangle\}$ for Hilbert space, a matrix eigenvalue problem arises,

$$A\mathbf{u} = E\mathbf{u}, \quad u_i = \langle\phi_i|\psi\rangle\,, \quad A_{ij} = \left\langle\phi_i|\hat{H}|\phi_j\right\rangle.$$

We prove the spectral theorem for normal operators over finite dimensional Hilbert spaces. The spectral theorem states that one can actually find a *basis* of eigenvectors for normal operators. The normal operators include the Hermitian operators.

In this section, we let $M(n) = M(n, n, \mathbb{C}) = \mathbb{C}^{n\times n}$ be the set of complex square matrices of dimension n.

Two matrices A and B are said to be *similar* if there is an invertible matrix U such that $A = UBU^{-1}$. If U is unitary, then A and B are *unitarily similar*. (U is unitary if and only if $U^H = U^{-1}$.)

Exercise 11.1.

1. What does it mean that $A \in M(n)$ is normal?
2. Prove that a diagonal matrix $D \in M(n)$ is normal. (A matrix A is diagonal if and only if $A_{ij} = 0$ whenever $i \neq j$.)
3. Prove that U is unitary if and only if its columns are orthonormal.
4. Prove that if A is normal and B is unitarily similar to A, then B is normal.
5. Prove that if A is unitarily equivalent to a diagonal matrix, then A is normal.

The next part requires *Schur's Lemma*:

> **Lemma: Schur's Lemma.** Let $A \in M(n)$. Then A is unitarily equivalent to an upper triangular matrix. (R is upper triangular if $R_{ij} = 0$ whenever $i > j$.)

The proof of Shur's Lemma can be found in, e.g., Fraleigh and Beauregard.

1. Prove that every normal matrix A is unitarily equivalent to a normal upper-triangular matrix B. (Use Schur's Lemma and exercise that normality is preserved by unitary similarity.)
2. Prove that a normal upper triangular matrix B must be diagonal. Hint: Let $C = B^H B = BB^H$, and compute C_{11} to show that $B_{1j} = 0$ if $j > 1$. Continue to C_{22}, and all the way up to C_{nn}.

We have now proved:

> **Theorem: Spectral theorem for normal matrices.** Suppose $A \in M(n)$ is normal, i.e., $AA^H = A^H A$. Then there is a unitary matrix $U \in M(n)$ and a diagonal matrix $D \in M(n)$ such that
>
> $$A = UDU^H. \quad \text{spectral decomposition}$$

1. Let $\mathbf{u}_i = U_{:,i}$ be the ith column for U. Explain why these vectors form a basis for $\mathbb{C}^n$.
2. Show that

   $$A = \sum_{i=1}^{n} d_i \mathbf{u}_i \mathbf{u}_i^H, \quad d_i = D_{ii}. \quad \text{spectral decomposition}$$

   Write this formula also using bra-ket notation.

3. What does A do with the vector $\mathbf{u}_k$ for some k? What does A do to a linear combination of the basis vectors?

A special kind of normal matrix is a Hermitian matrix, for which

$$A^H = A.$$

1. Show that a Hermitian matrix is unitarily equivalent to a diagonal matrix with *real* numbers on the diagonal, i.e., $A = UDU^H$ with $D_{ii} \in \mathbb{R}$.

11.1.2 Diagonalization of a matrix

A matrix A has an eigenvalue E if and only if there is a nonzero solution to the linear system

$$(A - EI)\mathbf{x} = 0.$$

Thus, the matrix $A - EI$ must be singular, i.e., not invertible. The eigenvector $\mathbf{x}$ is then a vector in the null space of this matrix.

A sufficient and necessary criterion for invertibility of a matrix B is that the *determinant* is nonzero. The definition of the determinant can be stated as follows:

> **Definition: Determinant.** Let $A \in M(n)$. The determinant, written $\det(A) \in \mathbb{C}$ is defined by
>
> $$\det(A) = \sum_{p \in S_n} (-1)^{|p|} A_{1,p(1)} A_{2,p(2)} \cdots A_{n,p(n)}.$$
>
> Here, S_N is the set of *permutations* of the numbers $\{1, 2, \cdots, n\}$. A permutation is by definition any bijective map.

Each permutation $p \in S_n$ is a rearrangement is thus completely specified by the new order of the numbers $\{1, 2, \cdots, n\}$, i.e., $(p(1), p(2), \cdots, p(n))$. For example, $(1, 2, 3, 4, 5, 6) \in S_6$ is the *identity permutation* that leaves any number alone. The permutation $(3, 2, 1, 4, 6, 5)$ makes 1 and 3 switch place, and 5 and 6.

Any permutation can be written as a product (composition) of *transpositions*, i.e., exchanges of exactly two elements. Thus, $(3, 1, 2) = (1, 2)(2, 3)$. Here, the two-element notation indicates which elements are switched. Every decomposition of a given permutation uses either an even number of transpositions or an odd number, denoted $|p|$ modulo 2. The *sign* of a permutation p is $(-1)^{|p|}$.

There are $n!$ unique permutations of an n-element set.

Exercise 11.2. Write out the set of permutations S_1, S_2 and S_3.

Exercise 11.3. Compute the sign of the permutations of the previous exercise.

Exercise 11.4. Compute explicit expressions for the determinants of matrices of dimension 1, 2, and 3.

Exercise 11.5. (Extra.) In Python, the module `itertools` contains an iterator class `permutations` that allow efficient looping over permutations. Can you write a function that computes the sign of a permutation? Use the internet for efficiency.

Exercise 11.6. Compute explicit expressions for the determinants of matrices of dimension up to 3. By equating the determinant of $A - EI$ with zero, a polynomial equation in E is obtained. Find the solutions in the case $n = 2$.

Exercise 11.7. Consider the matrix

$$A = \begin{bmatrix} -1 & z \\ z & 1 \end{bmatrix}.$$

For which $z \in \mathbb{C}$ is A Hermitian? Normal? Compute the eigenvalues as function of z. Also find the eigenvectors.

Exercise 11.8. For which values of z does there exist a basis of eigenvectors?

11.2 Solutions

11.2.1 Tutor note: Eigenvalue decomposition

Tutor note: This exercise may be considered for the especially interested student. It requires some mathematical argument skill. I would not consider it *hard*, though, but perhaps not focus on this exercise to begin with. I will try to make a solution for this in time.

Solution to Exercise 11.1

The source does not supply a solution.

11.2.2 Tutor note: Diagonalization of a matrix

Tutor note: This exercise, I find quite important for understanding eigenvalue problems. I therefore recommend it. The students could also use computations in Python or similar if they find the analytic part a little hard.

I think this is one of th exercises that the tutors should spend some time with.

Solution to Exercise 11.2

The source does not supply a solution.

Solution to Exercise 11.3

The source does not supply a solution.

Solution to Exercise 11.4

The source does not supply a solution.

Solution to Exercise 11.5

The source does not supply a solution.

Solution to Exercise 11.6

The source does not supply a solution.

Solution to Exercise 11.7

The source does not supply a solution.

Solution to Exercise 11.8

The source does not supply a solution.

12 Gram–Schmidt orthogonalization

Orthogonal projection leads to the Gram–Schmidt construction and the QR factorization of a full-rank matrix.

Let V be a vector space over $\mathbb{F}$ with $\dim(V) = n < +\infty$. Recall that every finite-dimensional vector space has a basis. In the lecture notes, it is claimed that every finite dimensional Hilbert space has a basis of orthonormal vectors! Recall, that in an orthonormal basis, computations are usually much simpler than in nonorthogonal bases.

In this exercise, we show that any set of linearly independent vectors can be orthonormalized. We work in the space $\mathbb{F}^n$ for simplicity. That is, let $B = [\mathbf{b}_1, \cdots, \mathbf{b}_m] \subset \mathbb{F}^n$ be a set of $m \leq n$ linearly independent vectors, spanning a subspace $V = \operatorname{span}(B)$. We denote the vector set by its basis matrix B. Equivalently, B is an arbitrary matrix in $M(n, m, \mathbb{F})$ of full rank m, and the vectors $\mathbf{b}_i$ are the columns of B.

We will find a set of orthonormal vectors $U = [\mathbf{u}_1, \cdots, \mathbf{u}_m]$, again denoted by a matrix in $M(n, m, \mathbb{F})$, such that $\operatorname{span}(B) = \operatorname{span}(U)$. That is, the two sets of vectors are bases for the *same* subspace, but one of them is an orthonormal basis.

We first define the notion of orthogonal projection onto the span of a single vector: Let $\mathbf{y} \in \mathbb{F}^n$ be nonzero, $Y = \operatorname{span}(\mathbf{y})$, and let $\mathbf{x} \in \mathbb{F}^n$.

12.1 Exercises

Exercise 12.1.

1. Show that the vector $\mathbf{x}_Y$ given by

$$\mathbf{x}_Y = \mathbf{y}\frac{\langle \mathbf{y}, \mathbf{x} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle}$$

is the *unique* vector in Y, such that

$$\forall \mathbf{y}' \in Y, \quad \|\mathbf{x} - \mathbf{x}_Y\| \leq \|\mathbf{x} - \mathbf{y}'\|.$$

Hint: Find a function $f : \mathbb{R} \to \mathbb{R}$ which you can study, by using the definition of $Y = \text{span}(\mathbf{y})$.

2. Show that

$$\mathbf{x} = \mathbf{x}_Y + \mathbf{x}_{Y^\perp},$$

where

$$\langle \mathbf{x}_{Y^\perp} | \mathbf{x}_Y \rangle = 0.$$

Illustrate with a picture of the situation in $\mathbb{R}^2$.

The vector $\mathbf{x}_Y$ is called the *orthogonal projection of* $\mathbf{x}$ *onto* Y *or* $\mathbf{y}$.

1. Explain that the orthogonal projection is given by acting with an operator,

$$\mathbf{x}_Y = P_{\mathbf{y}}\mathbf{x}, \quad P_{\mathbf{y}} = \mathbf{y} \langle \mathbf{y}, \mathbf{y} \rangle^{-1} \mathbf{y}^H.$$

The operator $P_{\mathbf{y}}$ is called the *orthogonal projection operator onto* Y = span($\mathbf{y}$).

1. Show that $P^2 = P$ and that $P^\dagger = P$.

The Gram–Schmidt orthogonalization proceeds recursively:

1. $\mathbf{u}_1 = \mathbf{b}_1 \langle \mathbf{b}_1, \mathbf{b}_1 \rangle^{-1/2}$ (normalization)
2. $\mathbf{u}_2' = \mathbf{b}_2 - P_{\mathbf{u}_1}\mathbf{b}_2$ (removal of projection), and $\mathbf{u}_2 = \mathbf{u}_2' \langle \mathbf{u}_2', \mathbf{u}_2' \rangle^{-1/2}$ (normalization)
3. $\mathbf{u}_3' = \mathbf{b}_3 - P_{\mathbf{u}_1}\mathbf{b}_3 - P_{\mathbf{u}_2}\mathbf{b}_3$ (removal of projection), and $\mathbf{u}_3 = \mathbf{u}_3' \langle \mathbf{u}_3', \mathbf{u}_3' \rangle^{-1/2}$ (normalization)
4. Continue recursively.
5. $\mathbf{u}_m' = \mathbf{b}_m - \sum_{i=1}^{m-1} P_{\mathbf{u}_i}\mathbf{b}_m$ (removal of projection), and $\mathbf{u}_m = \mathbf{u}_m' \langle \mathbf{u}_m', \mathbf{u}_m' \rangle^{-1/2}$ (normalization)

We see that each $\mathbf{u}_i$ is given by projecting away from $\mathbf{b}_i$ the components of all the *previously generated* $\mathbf{u}_j$, $j < i$.

1. Show that all the generated vectors $\mathbf{u}_i$ are orthonormal.
2. Show that a set of orthonormal vectors are linearly independent.

3. Show that the span of the first k original vectors is the same as the span of the first k generated vectors. Show that

$$B = UR,$$

where $R \in M(m, m, \mathbb{F})$ is *upper triangular*, i.e., $R_{ij} = 0$ whenever $i > j$. This is called the QR decomposition of the operator B. (The "Q" is just our U.)

4. We have shown the existence of the QR decomposition when B had full rank. Can you show the existence of the decomposition when B does *not* have full rank? It always exists. Hint: What happens when B does not have full rank?

We now turn to a computational exercise:

1. Write a Python function to compute the QR decomposition of a matrix with full rank. In SciPy, the function `scipy.linalg.qr` is an advanced implementation with pivoting. To compare with your implementation, use the call

```
Q,R = qr(A,mode='economic',pivoting=False)
```

With `pivoting=False`, the function computes essentially the same as our presented algorithm. (Turning on pivoting stabilizes the algorithm, rearranging the vectors before orthogonalizing.)

2. Let $n \geq m > 0$ be integers, and define the matrix $A \in \mathbb{R}^{n \times m}$ by

$$A_{ij} = \left(\frac{i-1}{n-1}\right)^{j-1}.$$

This the matrix of the m first monomials x^{j-1} evaluated at an equidistant grid of n points in the interval $[0, 1]$. Test your QR implementation on A for various n, m, say $m = 10$ and $m = 200$. In particular, compute the difference between your factorization and A, and compare also with the NumPy implementation.

12.2 Solutions

12.2.1 Tutor note: Gram–Schmidt orthogonalization

Tutor note: I think Gram–Schmidt is important, but maybe there is not enough time in the exercise session.

This exercise may be hard for students who have little experience with proofs. Tutors should prepare the projection and span arguments.

Solution to Exercise 12.1

For the rank-deficient QR question.

Sketch: When B has rank $m' < m$, remove $m - m'$ dependent columns and factor the reduced matrix. Extend the resulting orthonormal set with $m - m'$ vectors orthogonal to those already generated, and add the corresponding rows and columns to R so that $B = UR$ still holds.

13 The singular-value decomposition

The singular-value decomposition gives orthogonal matrix expansions and low-rank approximations, including image compression.

The SVD is a powerful linear algebra tool. In quantum chemistry it is used to compress electron-repulsion integrals, for example.

In this small exercise, we demonstrate its power applied to *image compression*. For this exercise, a good grayscale image is handy. You can find a nice picture of a parrot in Figure 13.1.

Recall first the *Hilbert–Schmidt* inner product on the set $M(n, m, \mathbb{F})$ of matrices:

$$\langle X, Y \rangle_{\mathrm{HS}} = \mathrm{Tr}(X^H Y)$$

The *trace* $\mathrm{Tr}(A)$ is the sum of the diagonal elements of A.

Recall also the SVD: For $A \in M(n, m, \mathbb{F})$ and $k = \min(n, m)$, there are unitary matrices $U \in M(n)$ and $V \in M(m)$ and singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \geq 0$ such that

$$A = U\Sigma V^H = \sum_{i=1}^{k} \mathbf{u}_i \sigma_i \mathbf{v}_i^H.$$

Here $\Sigma \in M(n, m, \mathbb{R})$ has the singular values in its upper $k \times k$ diagonal block and zeros elsewhere. Removing unused columns gives the economical SVD

$$A = U_{:,:k} \Sigma_{:k,:k} V^H_{:,:k}.$$

Figure 13.1: A parrot, https://www.dropbox.com/s/8r4svs6uuhgwawt/parrot.png

13.1 Exercises

Exercise 13.1.

1. How is the Hilbert–Schmidt inner product related to the Euclidean inner product and the standard Euclidean vector space?
2. Show that $\mathrm{Tr}(ABC) = \mathrm{Tr}(CAB)$ (cyclic invariance).
3. Let $\mathbf{u}_i$ and $\mathbf{v}_i$ be the left and right singular vectors of a matrix A. Show that the matrices $X^{ij} = \mathbf{u}_i\mathbf{v}_j^H$ form an orthonormal set in the Hilbert–Schmidt inner product.
4. Conclude that the SVD is an expansion of a matrix in an orthonormal basis.

The *truncated* SVD

$$A^{(m)} = \sum_{i=1}^{m} \mathbf{u}_i\sigma_i\mathbf{v}_i^H$$

is the best approximation to A of rank at most m in the Hilbert–Schmidt norm:

$$A^{(m)} = \operatorname*{arg\,min}_{\mathrm{rank}(B)\leq m} \|B - A\|_{\mathrm{HS}}.$$

1. Write a Python program/notebook that reads an image file and converts it to a matrix $A \in M(n, m, \mathbb{R})$. The module `imageio` can be useful.
2. Continue writing the program, so it computes the truncated and full SVD of A. You can use, say, `scipy.linalg.svd` to find the full SVD.
3. Plot the singular values. Discuss.
4. Show that the error in the Hilbert–Schmidt norm is

$$\|A^{(m)} - A\|_{\mathrm{HS}} = \sqrt{\sum_{i=m+1}^{k} \sigma_i^2}.$$

5. Make visualizations of the truncated SVD of the image A, with various m. Compute the relative error $\|A^{(m)} - A\|_{\mathrm{HS}}/\|A\|_{\mathrm{HS}}$ for each m you consider. Discuss the image quality.

13.2 Solutions

13.2.1 Tutor note: The singular value decomposition as a compression tool

Tutor note: This exercise combines computations with accessible analysis. It will probably take most of an hour, depending on the student.

Solution to Exercise 13.1

For the orthonormal-matrix question.

$$\left\langle X^{ij}, X^{i'j'}\right\rangle_{\mathrm{HS}} = \mathrm{Tr}(\mathbf{v}_j \mathbf{u}_i^H \mathbf{u}_{i'} \mathbf{v}_{j'}^H) = \langle \mathbf{u}_i, \mathbf{u}_{i'}\rangle \left\langle \mathbf{v}_j, \mathbf{v}_{j'}\right\rangle = \delta_{ii'}\delta_{jj'}.$$

14 Polynomial vector spaces

Polynomials form finite-dimensional vector spaces on which differentiation, multiplication, and changes of basis act as linear maps.

In this exercise, we study a vector space that is not $\mathbb{F}^n$. We start out without an inner product, but supply this later.

Recall that a polynomial of degree n over $\mathbb{F}$ has the form

$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n,$$

with $a_i \in \mathbb{F}$ and $a_n \neq 0$.

14.1 Exercises

Exercise 14.1. Let P_n be the set of all polynomials of degree less than or equal to n. Show that P_n is a vector space of dimension $n + 1$ over $\mathbb{C}$.

Exercise 14.2. Let $D : P_n \to P_n$ be the differentiation operator, i.e.,

$$(Dp)(x) = p'(x).$$

Show that D is indeed a linear operator.

Exercise 14.3. Does there exist an inverse of D?

Exercise 14.4. Let $B = \{1, x, x^2, \cdots, x^n\}$ be the basis of *monomials*. Using bra-ket notation, we write the basis operator as

$$\hat{B} = \sum_{i=1}^{n+1} \left|x^{i-1}\right\rangle \langle i| \,,$$

where $|i\rangle$ is a standard basis vector for $\mathbb{F}^{n+1}$. Compute the matrix of D relative to this basis. (Note carefully that we do not have an inner product defined.) That is, find a matrix T such that

$$D\hat{B} = \hat{B}T.$$

We now restrict our attention to the interval $x \in [-1, 1]$. That is,

$$V_n = \{p : [-1, 1] \to \mathbb{C} \mid p \text{ a polynomial of deg} \leq n\}.$$

The above subexercises did not depend on the domain of the polynomial.

We define an inner product on V,

$$\langle p|q\rangle := \int_{-1}^{1} \overline{p(x)} q(x) \, \mathrm{d}x.$$

We will now define the *Legendre polynomials.*

Exercise 14.5. Show that $\langle \cdot | \cdot \rangle$ is indeed an inner product by checking the axioms.

Exercise 14.6. Compute the overlap matrix S of the monomial basis. Is the basis orthonormal?

Exercise 14.7. Formulate the Gram–Schmidt procedure using the bra-ket notation. Explain how the procedure generates an upper triangular matrix R such that

$$\hat{B} = \hat{U} R,$$

where $\hat{U}$ is the basis matrix of the orthonormal set of vectors.

Exercise 14.8. Use the Gram–Schmidt decomposition by hand to compute the first three *normalized Legendre polynomials* $|\hat{P}_i\rangle$, $i = 0, 1, 2$, defined using the Gram–Schmidt orthogonalization of $\{1, x, x^2\}$.

The Legendre polynomials are orthogonal polynomials $P_i : [-1, 1] \to \mathbb{R}$ obtained from the monomials by Gram–Schmidt orthogonalization and normalized by $P_i(1) = 1$. They are orthogonal but not normalized. The endpoint normalization gives

$$\int_{-1}^{1} P_i(x) P_j(x) \, \mathrm{d}x = \frac{2}{2i + 1} \delta_{ij}.$$

With $P_0(x) = 1$ and $P_{-1} \equiv 0$, they satisfy the recurrence relation

$$(i + 1) P_{i+1}(x) = (2i + 1) x P_i(x) - i P_{i-1}(x).$$

From this, one can show:

$$P'_{i+1}(x) = \frac{2P_i(x)}{\|P_i\|^2} + \frac{2P_{i-2}(x)}{\|P_{i-2}\|^2} + \frac{2P_{i-4}(x)}{\|P_{i-4}\|^2} + \dots$$

Exercise 14.9. Compute the matrix of the differential operator D in the *normalized* Legendre polynomial basis, both by using matrix multiplication involving X, R, and R^{-1}, and also by using the derivative relation above. Compare the two explicitly using pen-and-paper calculations.

Exercise 14.10. Repeat the previous exercise with the operator D^2.

Orthogonal polynomials define efficient *quadrature rules*. Consider approximating an integral by

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx \sum_{i=1}^{n} f(x_i)w_i,$$

where the w_i are *weights* and the x_i are *nodes*. A unique choice of nodes and weights makes the rule exact for every polynomial of degree at most $2n-1$. The nodes are the zeros of P_n, and the weights are

$$w_i = \frac{2}{(1-x_i^2)[P_n'(x_i)]^2}.$$

The Wikipedia page on Gauss–Legendre quadrature contains more details.

The so-called Golub–Welsch algorithm is a simple algorithm based on diagonalization of a tridiagonal Hermitian matrix for computing the nodes and weights of Gaussian quadratures, see the Wikipedia page on Gaussian quadrature. The interested student should check it out!

14.2 Solutions

14.2.1 Tutor note: Space of polynomials

Tutor note: Treating functions as vectors is important in quantum chemistry. The recommended sub-exercises should be broadly accessible; the remaining exercises may require tutor preparation.

Solution to Exercise 14.1

The sum of two polynomials is a polynomial. A scalar multiple of a polynomial is a polynomial. The polynomials x^i are clearly linearly independent. Since all polynomials are linear combinations of such, the dimension must be equal to the number of coefficients, $n+1$.

Solution to Exercise 14.2

The source does not supply a solution.

Solution to Exercise 14.3

No inverse exists, since the null space of D is nontrivial (the constant polynomials). The indefinite integral would be a candidate for inverse, but the integral is only determined up to a constant.

Solution to Exercise 14.4

Since $(x^{i-1})' = (i-1)x^{i-2}$ for $i > 1$, we get

$$T_{i-1,i} = i - 1, \qquad i = 2, \dots, n+1,$$

and all other matrix elements are zero.

Solution to Exercise 14.5

The source does not supply a solution.

Solution to Exercise 14.6

$$\int_{-1}^{1} x^i x^j \, \mathrm{d}x = \frac{1}{i+j+1}(1 - (-1)^{i+j+1}).$$

Clearly not an orthonormal basis, since the matrix is not the identity matrix.

Solution to Exercise 14.7

The source does not supply a solution.

Solution to Exercise 14.8

The source does not supply a solution.

Solution to Exercise 14.9

The source does not supply a solution.

Solution to Exercise 14.10

The source does not supply a solution.

Part IV

Metric spaces and topology

15 Metric spaces

A metric measures distance through symmetry, positivity, and the triangle inequality.

A *metric* encodes the intuition behind measuring *distances* in a set M. It is a distance function $d(x, y)$, where $x, y \in M$.

The definition of a metric space is as follows:

> **Definition: Metric.** Let M be a set. A function $d : M \times M \to \mathbb{R}$ is a *metric* if it satisfies the following axioms:
>
> 1. $d(x, y) = d(y, x)$
> 2. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$
> 3. $d(x, y) \leq d(x, z) + d(z, y)$
>
> The pair (M, d) is a *metric space*. If M is a vector space, we say that (M, d) is a *metric vector space*.

In this exercise, we will try to understand why the axioms are as they are, i.e., what makes them encode the concept of a metric.

15.1 Exercises

Exercise 15.1.

1. Consider two persons X and Y located at x and y in a set M. They both measure the distance to the other person. What would happen if axiom a) is violated? Is this reasonable?
2. What would happen if axiom b) is violated? Is this reasonable?
3. A third person Z located at z enters the picture. Consider axiom c). What happens if this axiom is violated? Is this reasonable?

4. Let $M = \mathbb{R}^2$, and let d be Euclidean distance, "as the crow flies". But there are other distance measures as well. Consider for example the "Manhattan distance", given by

$$d(\mathbf{x}, \mathbf{y}) = |x_1 - y_1| + |x_2 - y_2|.$$

   This has a standard interpretation in terms of cities with straight streets and taxis driving on these streets. Discuss.

5. Consider a king moving on a chessboard. The king can move one step at a time, either horizontally or vertically, or diagonally. Consider the *minimum number of steps* the king has to move in order to get from one point to another. See Figure 15.1 for an illustration. The metric is

$$d(\mathbf{x}, \mathbf{y}) = \max(|x_1 - y_1|, |x_2 - y_2|).$$

   This is also called the "maximum metric". Can you verify the metric axioms? How does the maximum metric compare to the Manhattan metric?

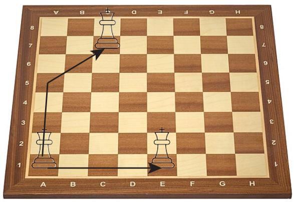


Figure 15.1: Illustration of "chessboard metric": The king can move horizontally, vertically, and diagonally. The chessboard metric is the smallest number of steps the king must move in order to get from one point to another. Here, the king starts at A1, and two paths are shown: The optimal path to C7, and the optimal path to E1. Note that the king will move diagonally as much as he can, because he travels "faster" that way!

15.2 Solutions

Solution to Exercise 15.1

For part 1.

The two observers would disagree about the distance between the same pair of points. A directed cost can be asymmetric, but it is not a metric under this definition.

For part 2.

Zero distances between non-equal points violate the concept of distances measuring how far apart points are.

For part 3.

The triangle inequality embodies an essential feature of Euclidean space: Taking detours should never lead to shorter distances traveled!

For part 5.

One possible set of observations is the following: The maximum metric is smaller than the Manhattan metric. On the other hand, the Manhattan metric is smaller than 2 times the maximum metric. This means that open sets in the Manhattan metric will be open sets in the maximum metric, and vice versa, since inside an open ball for each metric, we can always place an open ball of the other metric. Therefore, they create the same topology. A function $f : \mathbb{R}^2 \to \mathbb{R}$ is continuous with respect to one metric iff it is continuous with respect to the other metric. (The same is also true for the Euclidean metric, they are all *equivalent*.)

16 Open and closed sets in Euclidean space

Open balls define interior points, open sets, closed sets, boundaries, and closures in Euclidean space.

16.1 Exercises

Exercise 16.1. Show that any ϵ-ball is open.

Exercise 16.2. Show that $A \cup \partial A$ is closed.

Exercise 16.3. Show that if $A \subset \mathbb{R}^n$ is closed and a sequence $\mathbf{x}_i \in A$ converges to some $\mathbf{x} \in \mathbb{R}^n$, then $\mathbf{x} \in A$. This gives an equivalent characterization of closed subsets of $\mathbb{R}^n$.

Exercise 16.4. Show that the closure of A, the smallest closed set $\bar{A}$ that contains A, is equal to $A \cup \partial A$.

Exercise 16.5. Show that

$$A = \{(x, 0) \mid x \in [-1, 1]\} \subset \mathbb{R}^2$$

is closed.

Exercise 16.6. Show that $\mathbb{Q} \subset \mathbb{R}$ is neither open nor closed. Show that the boundary of $\mathbb{Q}$ is $\mathbb{R}$. Show that the interior of $\mathbb{Q}$ is empty. What is the closure of $\mathbb{Q}$?

Exercise 16.7. Let

$$A = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x < 1, \quad 0 \leq y < 1\}.$$

Compute the interior of A, the boundary of A, the closure of A.

16.2 Solutions

16.2.1 Tutor note: Open and closed sets in Euclidean space

Tutor note: This exercise really develops the basic skills thinking about open and closed sets, and also mathematical argumentation. Even if the exercises are fairly basic by undergrad maths standards, maybe this style of thinking is difficult for some students, and therefore I think it is worthwhile.

Solution to Exercise 16.1

Let $A = B_\epsilon(\mathbf{x}_0)$ and $\mathbf{x} \in A$. Set $r = \|\mathbf{x} - \mathbf{x}_0\| < \epsilon$ and $\delta = \epsilon - r > 0$. If $\mathbf{y} \in B_\delta(\mathbf{x})$, then

$$\|\mathbf{y} - \mathbf{x}_0\| \le \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x} - \mathbf{x}_0\| < \delta + r = \epsilon.$$

Thus $B_\delta(\mathbf{x}) \subset A$. Every point of A has an open ball contained in A, so A is open.

Solution to Exercise 16.2

Let $C = (A \cup \partial A)^\complement$. If $\mathbf{x} \in C$, then $\mathbf{x} \notin A$ and $\mathbf{x} \notin \partial A$. Since $\mathbf{x}$ is not a boundary point, some ball $B_\epsilon(\mathbf{x})$ lies wholly in A or wholly in $A^\complement$. It cannot lie in A because it contains $\mathbf{x}$, so it lies in $A^\complement$ and contains no boundary point. Hence $B_\epsilon(\mathbf{x}) \subset C$. The complement C is open, so $A \cup \partial A$ is closed.

Solution to Exercise 16.3

The source does not supply a completed solution.

Solution to Exercise 16.4

We have shown that $A \cup \partial A$ is closed. Suppose $A \cup \partial A$ is not the smallest closed set that contains A, i.e., $(A \cup \partial A) \setminus \bar{A}$ contains some point $\mathbf{x}$. Since both sets contain A, we must have $\mathbf{x} \in \partial A$. From this, we should look for a contradiction, because intuitively, the boundary must be part of $\bar{A}$. For all $\epsilon > 0$, $B_\epsilon(\mathbf{x})$ intersects both A and $A^\complement$. Therefore, there exists a sequence $\mathbf{x}_i \in A$ such that $\mathbf{x}_i \to \mathbf{x} \in \partial A$, but the smallest closed set containing A must contain this $\mathbf{x}$.

Solution to Exercise 16.5

Consider $A^\complement$, the plane without the interval $[-1, 1]$ on the x-axis. For a point $(x, y) \in A^\complement$ with $y \neq 0$, choose any $0 < \epsilon < |y|$. If $y = 0$, then $|x| > 1$, and we may choose any $0 < \epsilon < |x| - 1$. In either case, the ϵ-ball around (x, y) misses A. Thus $A^\complement$ is open and A is closed.

Solution to Exercise 16.6

Let $p/q \in \mathbb{Q}$ be a rational number, and let $]p/q - \epsilon, p/q + \epsilon[$ be an ϵ-"ball" around p/q. But we know that between any two real numbers, there are infinitely many rational numbers, and also irrational numbers. Hence, the ball is not contained in $\mathbb{Q}$, so the rational numbers do not form an open set. Hence no rational number has an open ball around it, and hence the interior of $\mathbb{Q}$ is empty. To show that $\mathbb{Q}$ is not closed, consider the openness of the complement, $\mathbb{R} \setminus \mathbb{Q}$, the irrational numbers. The same argument can be applied, the irrationals are not open.

The boundary of $\mathbb{Q}$ is all of $\mathbb{R}$, since any ball around a rational number contains both rational and irrational numbers.

Finally, the closure of $\mathbb{Q}$ is then $\mathbb{R}$.

Solution to Exercise 16.7

The interior of A is

$$\{(x, y) \in \mathbb{R}^2 \mid 0 < x < 1, \quad 0 < y < 1\}.$$

The boundary of A is the border of the square,

$\partial A = \{(0,t) \mid 0 \leq t \leq 1\} \cup \{(1,t) \mid 0 \leq t \leq 1\} \cup \{(t,0) \mid 0 \leq t \leq 1\} \cup \{(t,1) \mid 0 \leq t \leq 1\}$

The closure of A is A with the border, hence

$$\{(x,y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, \quad 0 \leq y \leq 1\}.$$

Part V

Calculus

Part VI

Multivariable calculus

17 Visualizing functions

Graphs, level sets, and vector fields provide complementary views of scalar- and vector-valued functions.

Some paths and their curves:

17.1 Exercises

Exercise 17.1. Let $f : \mathbb{R} \to \mathbb{R}^2$ be a path defined by

$$f(t) = [r(t)\cos(4t), r(t)\sin(4t)], \quad r(t) = \exp(-t).$$

Sketch the curve traced by the path for $0 \le t \le 2\pi$. You can check your sketch against, say, a Python plot.

Exercise 17.2. Let $f : (0, +\infty) \to \mathbb{R}^2$ be a path defined by

$$f(t) = [\ln(t), t].$$

Sketch the curve traced by the path. You can check your sketch against, say, a Python plot.

Exercise 17.3. Let $f : [0, \pi] \to \mathbb{R}^2$ be given by $f(t) = [x(t), y(t)]$, with $x(t) = 16\sin(t)^3$ and $y(t) = 13\cos(t) - 5\cos(2t) - 2\cos(3t) - \cos(4t)$. Sketch the curve. Here, it is probably easiest to use Python or a similar tool.

Level curves:

Exercise 17.4. Let $f(x, y) = x^2 + y^2$. Sketch the level curves $\{(x, y) \mid f(x, y) = n\}$, for $n = 1$ and $n = 2$, $n = 3$, and $n = 4$.

Exercise 17.5. Let $f(x, y) = x + y + 2$. Sketch the level curves where $f(x, y) = 0$, 2 and 4. Can you sketch the graph of f?

Exercise 17.6. Let $f(x,y) = x^2 - y^2$. The graph is called the *hyperbolic paraboloid*, or a saddle. Sketch the graph. Sketch the level curves $\{(x,y) \mid f(x,y) = n\}$, for $n = 0$ and $n = -1$, $n = 1$. Make sure that you get all "pieces".

Exercise 17.7. Write a Python program for sketching the level curves in the hyperbolic-paraboloid exercise above, but use more closely spaced constants. There are tools in Matplotlib that are handy.

Exercise 17.8. Adapt your program to draw level curves of $(x,y) \mapsto (x^2 + 3y^2)e^{1-x^2-y^2}$.

Sections and level surfaces:

Consider the hydrogen-atom eigenfunctions in atomic units. They are indexed by $n \geq 1$, $0 \leq \ell < n$, and $-\ell \leq m \leq \ell$. Their separated form is

$$\psi_{n\ell m}(r,\theta,\phi) = R_{n\ell}(r)Y_{\ell m}(\theta,\phi),$$

with

$$R_{n\ell}(r) = N_{n\ell}\rho^{\ell}L_{n-\ell-1}^{2\ell+1}(\rho)e^{-\rho/2}, \qquad \rho = \frac{2r}{n}.$$

Here $N_{n\ell}$ is a normalization constant. Coordinate conversion uses

$$r = \sqrt{x^2 + y^2 + z^2}, \quad x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta.$$

In particular, the $1s$ and $2p_z$ functions are given by

$$\psi_{1s}(x,y,z) = \frac{1}{\sqrt{\pi}}e^{-r(x,y,z)},$$

$$\psi_{2p_z}(x,y,z) = \frac{1}{4\sqrt{2\pi}}ze^{-r(x,y,z)/2},$$

Exercise 17.9. Write a Python program to draw the level *surfaces* of hydrogen orbitals, $\{(x,y,z) \mid \psi(x,y,z) = c\}$. Choose interesting values of c. It can be interesting to color code surfaces of opposite positive and negative c.

Exercise 17.10. Write a program to visualize *sections* of the orbital graphs. For example, fix $y = y_{\text{const}}$ and plot $(x,z) \mapsto \psi(x, y_{\text{const}}, z)$.

17.2 Solutions

17.2.1 Tutor note: Visualization of functions

Tutor note: The first few exercises are not hard, and the plots give quick visual feedback. The later exercises require some straightforward programming. The level-curve exercise comes from Marsden and Tromba.

Solution to Exercise 17.1

The source does not supply a solution.

Solution to Exercise 17.2

The source does not supply a solution.

Solution to Exercise 17.3

The source does not supply a solution.

Solution to Exercise 17.4

The source does not supply a solution.

Solution to Exercise 17.5

The source does not supply a solution.

Solution to Exercise 17.6

The source does not supply a solution.

Solution to Exercise 17.7

The source does not supply a solution.

Solution to Exercise 17.8

The source does not supply a solution.

Solution to Exercise 17.9

The source does not supply a solution.

Solution to Exercise 17.10

The source does not supply a solution.

18 Continuity and differentiability

Limits define continuity, while partial derivatives describe local change for functions of several variables.

18.1 Exercises

18.1.1 Continuity

Exercise 18.1. Let $f : \mathbb{R} \to \mathbb{R}$ be given by

$$f(x) = \begin{cases} 1 & x > 0 \\ -1 & x \leq 0 \end{cases}$$

For which $x \in \mathbb{R}$ does the limit

$$\lim_{h \to 0} f(x+h)$$

not exist? Why?

Exercise 18.2. Show that any polynomial $p : \mathbb{R} \to \mathbb{R}$ is continuous, by using the theorem on properties of continuous functions.

Exercise 18.3. (Marsden and Tromba) Consider the function

$$f(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}.$$

This function is defined on $\mathbb{R}^2 \setminus \{(0,0)\}$. Determine whether $f(x, y)$ approaches some value as $(x, y) \to (0, 0)$. What is this value?

Exercise 18.4. (Marsden and Tromba) Does $\lim_{(x,y)\to(0,0)} x^2/(x^2 + y^2)$ exist? If you want, plotting the function can help.

Exercise 18.5. (Marsden and Tromba) Prove that $\lim_{(x,y)\to(0,0)} 2x^2y/(x^2+y^2) = 0$ using an ϵ-δ argument.

18.1.2 Differentiability

Exercise 18.6. Compute the partial derivatives of

$$f(x,y) = \frac{xy}{(x^2+y^2)^{1/2}}.$$

18.2 Solutions

18.2.1 Tutor note: Continuity

Tutor note: The concept of continuity is basic, so I recommend some of these exercises for everyone. I do not think the tutors need much preparation for them.

Solution to Exercise 18.1

The limit does not exist at $x = 0$. For every interval $]-\epsilon, \epsilon[$, f maps into $\{-1, 1\}$, which does not approach a single value.

Solution to Exercise 18.2

The identity function $x \mapsto x$ is continuous. Since sums and products of continuous functions are continuous, every polynomial is continuous.

Solution to Exercise 18.3

Use $\lim_{t\to 0} \sin(t)/t = 1$. Given $\epsilon > 0$, choose $\eta > 0$ such that $0 < |t| < \eta$ implies $|\sin(t)/t - 1| < \epsilon$. Set $\delta = \sqrt{\eta}$. If $0 < \|\mathbf{u}\| < \delta$, then $0 < \|\mathbf{u}\|^2 < \eta$, so

$$\left| \frac{\sin(\|\mathbf{u}\|^2)}{\|\mathbf{u}\|^2} - 1 \right| < \epsilon.$$

Therefore the limit is 1.

Solution to Exercise 18.4

A hint to give the students:

Consider the line $x = 0$ and the line $y = 0$.

The limit does not exist. Along $y = 0$ the function equals 1, while along $x = 0$ it equals 0.

Solution to Exercise 18.5

We have

$$|2x^2y/(x^2 + y^2)| \leq |2x^2y/x^2| \leq 2|y|.$$

Thus, given $\epsilon > 0$, choose $\delta = \epsilon/2$. Then $0 < \|(x, y) - (0, 0)\| = \sqrt{x^2 + y^2} < \delta$ implies $|y| < \delta$, and so $|f(x, y) - 0| < 2\delta < \epsilon$.

18.2.2 Tutor note: Differentiability

Tutor note: The source contains only one exercise on differentiability; more exercises may be added later.

Solution to Exercise 18.6

For $(x, y) \neq (0, 0)$, write $v = (x^2 + y^2)^{1/2}$. The quotient and chain rules give

$$\frac{\partial f}{\partial x} = \frac{yv - xy(x/v)}{v^2} = \frac{y^3}{(x^2 + y^2)^{3/2}},$$

and, by symmetry,

$$\frac{\partial f}{\partial y} = \frac{x^3}{(x^2 + y^2)^{3/2}}.$$

Part VII

Series and matrix functions

19 Matrix exponentials and commutators

Power series define matrix exponentials. Differentiating products of exponentials produces nested commutators.

Consider the *matrix-valued function*

$$f(t) = e^{-tA} B e^{tA},$$

where A and B are square matrices. The matrix exponential is defined in terms of the Taylor series,

$$\exp(X) = \mathbf{1} + X + \frac{1}{2!}X^2 + \frac{1}{3!}X^3 + \cdots,$$

which is convergent. It is a fact that if $XY - YX = [X, Y] = 0$ (commuting matrices) then $\exp(X + Y) = \exp(X)\exp(Y)$, by the same proof as for the exponential function over a field $\mathbb{F}$.

19.1 Exercises

Exercise 19.1. Explain why we may view f as a vector-valued function

Exercise 19.2. Show that $t \mapsto e^{tA}$ is differentiable.

Exercise 19.3. Show that f is differentiable. Compute its derivatives to arbitrary order

19.2 Solutions

19.2.1 Tutor note: Taylor polynomials

Tutor note: I think it is important in quantum chemistry to be able to view matrices as vectors and as function parameters. On the other hand, some of this exercise may be hard for the student.

The first source item supplied the setup but no question, so it now appears in the chapter introduction.

Solution to Exercise 19.1

Identification of $\mathbb{F}^{n\times m}$ with $\mathbb{F}^{nm}$.

Solution to Exercise 19.2

Since tA and hA commute,

$$e^{(t+h)A} = e^{tA}e^{hA} = e^{tA}\left(\mathbf{1} + hA + O(h^2)\right).$$

Therefore

$$\lim_{h\to 0} \frac{e^{(t+h)A} - e^{tA}}{h} = e^{tA}A = Ae^{tA}.$$

Solution to Exercise 19.3

f is a product of differentiable matrix-valued functions. The product rule gives

$$f'(t) = [f(t), A].$$

Repeated differentiation gives

$$f^{(n)}(t) = \underbrace{[\cdots[[f(t), A], A], \dots, A]}_{n \text{ commutators}}.$$

Part VIII

Complex numbers and complex analysis

20 Complex arithmetic and geometry

Cartesian and polar forms connect complex arithmetic with plane geometry, rotations, dot products, and cross products.

20.1 Exercises

20.1.1 Complex numbers

Exercise 20.1. Let $z = x + \mathrm{i}y \in \mathbb{C}$, and compute the real and imaginary parts of z^2, z^3, and z^4, as functions of x and y.

Exercise 20.2. Compute a closed-form expression for the real and imaginary parts of z^n.

Exercise 20.3. Compute closed-form expression for z^{-1}, exhibiting the real and imaginary parts as functions of x and y. (Assuming $z \neq 0$). Try to compute closed-form expressions of z^{-2} and z^n as well.

Exercise 20.4. A complex number $z = x + \mathrm{i}y$ can be written in polar form,

$$x = r\cos(\theta), \quad y = r\sin(\theta).$$

Write down the rule for $z^n = R\cos\phi + \mathrm{i}R\sin\phi$ using angle θ and modulus r.

Exercise 20.5. Consider the polynomial equation $f(z) = z^3 + 1 = 0$. Find all the roots in the complex plane using polar coordinates. Sketch the roots in a coordinate system.

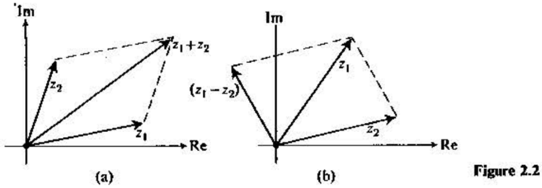


Figure 20.1: The parallelogram rule. (From Butkov.)

20.1.2 Basic calculations

Exercise 20.6. Let $z = x + \mathrm{i}y \in \mathbb{C}$. Illustrate the addition $w = z + \bar{z}$ in a coordinate system using the parallelogram rule.

Exercise 20.7. Recall our earlier exercise, where $\mathbb{C}$ was regarded as the real vector space $\mathbb{R}^2$. Let $z_i = x_i + \mathrm{i}y_i$, and denote the corresponding vectors in $\mathbb{R}^2$ by $\mathbf{v}_i$. Show that

$$\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = \Re(\overline{z_1} z_2) = \Re(z_1 \overline{z}_2).$$

Exercise 20.8. Another product in $\mathbb{R}^2$ (and indeed $\mathbb{R}^3$) is the *cross product*:

$$\mathbf{v}_1 \times \mathbf{v}_2 = \begin{vmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{vmatrix},$$

the matrix determinant. Show that

$$\mathbf{v}_1 \times \mathbf{v}_2 = \Im(\overline{z_1} z_2) = -\Im(z_1 \overline{z}_2).$$

Exercise 20.9. Show that the function $z \mapsto \mathrm{i}z$ is a counterclockwise rotation by $\pi/2$. Find the function that performs a clockwise rotation by the same angle.

20.2 Solutions

Solution to Exercise 20.1

$$(x + \mathrm{i}y)^2 = x^2 + 2\mathrm{i}xy + \mathrm{i}^2 y^2 = (x^2 - y^2) + 2\mathrm{i}xy.$$

Obtain the next powers by multiplying the preceding result by z.

Solution to Exercise 20.2

The binomial expansion gives

$$(x + \mathrm{i}y)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} (\mathrm{i}y)^j.$$

Even powers satisfy $\mathrm{i}^{2k} = (-1)^k$. The even indices between 0 and n are $2j$ for $j = 0, 1, \dots, \lfloor n/2 \rfloor$, while the odd indices are $2j + 1$ for $j = 0, 1, \dots, \lfloor (n-1)/2 \rfloor$. Let $K = \lfloor n/2 \rfloor$ and $L = \lfloor (n-1)/2 \rfloor$.

Splitting into even and odd terms gives

$$(x + \mathrm{i}y)^n = \sum_{j=0}^{K} (-1)^j \binom{n}{2j} x^{n-2j} y^{2j} + \mathrm{i} \sum_{j=0}^{L} (-1)^j \binom{n}{2j+1} x^{n-2j-1} y^{2j+1}.$$

Solution to Exercise 20.3

Multiply by the complex conjugate:

$$z^{-1} = \frac{1}{x + \mathrm{i}y} = \frac{x - \mathrm{i}y}{(x + \mathrm{i}y)(x - \mathrm{i}y)} = \frac{x - \mathrm{i}y}{x^2 + y^2}.$$

The formulas from the previous exercise then apply with $y \mapsto -y$ and a prefactor $(x^2 + y^2)^{-n}$.

Solution to Exercise 20.4

The geometric interpretation gives $\phi = n\theta$ (modulus 2π) and $R = r^n$.

Solution to Exercise 20.5

-1 has modulus 1 and angle π. A root must be such that $r^n = 1$ (which implies $r = 1$), and $3\theta = \pi + 2k\pi$, since the angle wraps around at 2π when we multiply complex numbers. Thus, $\theta = \pi/3 + 2k\pi/3$. $k = 0$ gives $\pi/3$, $k = 1$ gives $3\pi/3 = \pi$, and $k = 2$ gives $5\pi/3$. Now any larger k wraps around to these three angles. So we have found the roots.

Solution to Exercise 20.6

The source does not supply a solution.

Solution to Exercise 20.7

The source does not supply a solution.

Solution to Exercise 20.8

The source does not supply a solution.

Solution to Exercise 20.9

The source does not supply a solution.

21 Subsets of the complex plane

The modulus and real and imaginary parts describe curves and regions in the complex plane.

Visualize the following subsets of $\mathbb{C}$. Decide which sets are open, closed, or neither.

21.1 Exercises

Exercise 21.1. $\{z \mid |z-1| = 1\}$

Exercise 21.2. $\{z \mid |\mathrm{i}z-1| < 1\}$

Exercise 21.3. $\{z \mid |z-\mathrm{i}| + |z+\mathrm{i}| < 4\}$

Exercise 21.4. $\{z^2 \mid |z-\mathrm{i}| = 1\}$

Exercise 21.5. $\{z \mid \Re(e^{\mathrm{i}\pi/2}z) > 0\}$

Exercise 21.6. $\{z \mid \Im(z^2) > 0\}$

21.2 Solutions

Solution to Exercise 21.1

Square the equation and introduce x and y to get the equation for a circle, i.e., boundary of a circle. Alternatively just realize that the modulus $|w|$ is the distance from the origin.

Solution to Exercise 21.2

Interior of a circle.

Solution to Exercise 21.3

Ellipse with foci at $\pm\mathrm{i}$.

Solution to Exercise 21.4

Parameterize the original circle by $z(t) = \mathrm{i} + e^{\mathrm{i}t}$, $0 \leq t < 2\pi$, and plot or analyze the image curve $z(t)^2$.

Solution to Exercise 21.5

This is a half-plane.

Solution to Exercise 21.6

The source does not supply a solution.

22 The complex exponential

The functional equation for the exponential and the Cauchy–Riemann equations determine the complex exponential.

In this section, we give some relatively easy exercises to illustrate the important concepts of complex analytic functions.

In this exercise, we derive the complex exponential function. The starting point is the real exponential function, which is the unique function $f(x) = \exp(x)$ that satisfies

$$f(x_1 + x_2) = f(x_1)f(x_2), \quad f(0) = 1.$$

We desire to generalize $\exp(x)$ to the complex plane, so that $\exp(x + 0\mathrm{i}) = \exp(x) \in \mathbb{R}$.

To avoid confusion, we call this unknown function $f : \mathbb{C} \to \mathbb{C}$.

22.1 Exercises

Exercise 22.1. Show that $f(x+\mathrm{i}y) = \exp(x)f(\mathrm{i}y)$. (Thus, we need to determine $f(\mathrm{i}y)$.)

Exercise 22.2. We set $f(\mathrm{i}y) = A(y) + \mathrm{i}B(y)$. Show that

$$A(y) = B'(y), \quad B(y) = -A'(y),$$

and hence that $A''(y) = -A(y)$. Hint: Cauchy–Riemann.

Exercise 22.3. The general solution to the ODE $A'' = -A$ is

$$A(y) = \alpha\cos(y) + \beta\sin(y),$$

where α and β are constants to be determined. Show that $\alpha = 1$ and $\beta = 0$ are the only constants compatible with $f(z)$ being a generalization of the real exponential function.

Exercise 22.4. Conclude that

$$f(x + \mathrm{i}y) = e^x(\cos y + \mathrm{i} \sin y)$$

Exercise 22.5. Show that $(\exp(z))' = \exp(z)$, using Cauchy–Riemann, and that $\exp(z)$ is everywhere analytic.

Exercise 22.6. Show that the equation

$$\exp(z) = w$$

has infinitely many solutions for any $w \neq 0$.

Exercise 22.7. Show that $\exp(z) \neq 0$.

22.2 Solutions

22.2.1 Tutor note: The complex exponential function

Tutor note: I think that this exercise is very nice. It is not too hard if the student has some experience with the equation $y'' = -y$.

Solution to Exercise 22.1

The source does not supply a solution.

Solution to Exercise 22.2

The source does not supply a solution.

Solution to Exercise 22.3

The source does not supply a solution.

Solution to Exercise 22.4

The source does not supply a solution.

Solution to Exercise 22.5

The source does not supply a solution.

Solution to Exercise 22.6

The source does not supply a solution.

Solution to Exercise 22.7

The source does not supply a solution.

23 Complex power series

Geometric and exponential series give power-series representations of elementary complex functions.

23.1 Exercises

Exercise 23.1. Consider the geometric series

$$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + \ldots \longrightarrow \frac{1}{1-z}.$$

Show that the Nth partial sum is

$$\sum_{n=0}^{N} z^n = \frac{1-z^{N+1}}{1-z}.$$

Polynomial long division may be useful.

Exercise 23.2. Find an upper bound for $|z|$ such that the partial sum converges to $1/(1-z)$.

Exercise 23.3. The Taylor series for the exponential function is

$$\sum_{n=0}^{\infty} \frac{1}{n!} z^n.$$

Using the definition of the complex exponential, find the Taylor series for $\cos(x)$ and $\sin(x)$, for $x \in \mathbb{R}$.

Exercise 23.4. Make plots of the partial sums of the sine and cosine Taylor series to verify your result.

Exercise 23.5. Find the Taylor series for the function $f(z) = (e^z - 1)/z$

Exercise 23.6. Find the Taylor series for the function $f(z) = \sin(z)/z$ for $z \neq 0$ and $f(0) = 1$.

23.2 Solutions

Solution to Exercise 23.1

The source does not supply a solution.

Solution to Exercise 23.2

The source does not supply a solution.

Solution to Exercise 23.3

The source does not supply a solution.

Solution to Exercise 23.4

The source does not supply a solution.

Solution to Exercise 23.5

The source does not supply a solution.

Solution to Exercise 23.6

The source does not supply a solution.

24 Singularities and contour integrals

Laurent series classify isolated singularities and determine contour integrals around poles.

24.1 Exercises

24.1.1 Singularities

Exercise 24.1. Consider the function $f(z) = \frac{1}{z(z-1)}$. How many poles does f have, and where are they? What are the order of the poles? Can you write down the Laurent series of f around $z = 0$?

Exercise 24.2. Does the function $f(z) = 1/\sin(z)$ have a pole? Where? What order?

Exercise 24.3. Find the singularities of $f(z) = e^{-1/(z-1)^2}$.

24.1.2 Line integrals

Let

$$f(z) = \frac{1}{z^3 - (2+\mathrm{i})z^2 + (1+2\mathrm{i})z - \mathrm{i}}.$$

Exercise 24.4. Let $\Gamma_\epsilon(w)$ be the simple closed curve defined by the counter-clockwise border of the ϵ-ball $B_\epsilon(w)$. Write down a parameterization (a path) for this curve.

Exercise 24.5. Identify the poles $P = \{w_1, w_2, \cdots\}$ of $f(z)$ and their order. Hint: A zero of the denominator is $z = \mathrm{i}$.

Exercise 24.6. What is the value of

$$\oint_{\Gamma_\epsilon(0)} f(z)\,\mathrm{d}z$$

for $\epsilon = 1/2$?

Exercise 24.7. Find the value of the line integrals

$$\oint_{\Gamma_\epsilon(w_i)} f(z)\,\mathrm{d}z$$

for $\epsilon = 1/2$. Hint: Do not attempt the integral directly; compute the relevant Laurent coefficient. It can be useful to use the Taylor expansion

$$\frac{1}{a-z} = \frac{1}{a}\frac{1}{1-z/a} = \frac{1}{a}\sum_{n=0}^{\infty}(z/a)^n.$$

24.2 Solutions

Solution to Exercise 24.1

The poles are simple and occur at $z = 0$ and $z = 1$. For $0 < |z| < 1$,

$$f(z) = \frac{1}{z(z-1)} = -\frac{1}{z(1-z)} = -z^{-1} - 1 - z - z^2 - \cdots.$$

Solution to Exercise 24.2

The zeros of $\sin z$ occur at $z = k\pi$, $k \in \mathbb{Z}$, and each zero is simple because $\cos(k\pi) = (-1)^k \neq 0$. Thus $1/\sin z$ has a simple pole at every $k\pi$. Near the origin, $1/\sin z \sim 1/z$.

Solution to Exercise 24.3

The exponent has a pole at $z = 1$ (of order 2). This means that the exponential function gives an essential singularity at $z = 1$.

Solution to Exercise 24.4

$\gamma(t) = w + \epsilon e^{it}$, $0 \le t < 2\pi$.

Solution to Exercise 24.5

The denominator can be factorized as $(z-1)^2(z-\mathrm{i})$, and therefore we have a pole of order 2 at $w_1 = 1$ and a pole of order 1 at $w_2 = \mathrm{i}$.

Solution to Exercise 24.6

The function has no poles inside the ϵ-ball, so the integral is zero by Cauchy's integral theorem.

Solution to Exercise 24.7

Since

$$f(z) = \frac{1}{(z-1)^2(z-\mathrm{i})},$$

the residue at the second-order pole $z = 1$ is

$$\operatorname{Res}(f, 1) = \frac{\mathrm{d}}{\mathrm{d}z} \frac{1}{z-\mathrm{i}}\bigg|_{z=1} = -\frac{1}{(1-\mathrm{i})^2} = -\frac{\mathrm{i}}{2}.$$

The residue at the simple pole $z = \mathrm{i}$ is

$$\operatorname{Res}(f, \mathrm{i}) = \frac{1}{(\mathrm{i}-1)^2} = \frac{\mathrm{i}}{2}.$$

Therefore

$$\oint_{\Gamma_{1/2}(1)} f(z)\,\mathrm{d}z = \pi, \qquad \oint_{\Gamma_{1/2}(\mathrm{i})} f(z)\,\mathrm{d}z = -\pi.$$

Part IX

Numerical methods

25 Dual numbers and automatic differentiation

Dual numbers adjoin a nonzero element ε with $\varepsilon^2 = 0$, which encodes first derivatives algebraically.

We now consider the number system called *the dual numbers*, often written $\mathbb{R}(\varepsilon)$. The dual numbers form an algebraic extension of $\mathbb{R}$ with a nonzero element ε such that $\varepsilon^2 = 0$:

$$\mathbb{R}(\varepsilon) = \{x + \varepsilon y \mid x, y \in \mathbb{R}\}.$$

Their construction resembles $\mathbb{C} = \mathbb{R}(\mathrm{i})$, but $\mathrm{i}^2 = -1$ while $\varepsilon^2 = 0$. Dual numbers support automatic differentiation.

25.1 Exercises

Exercise 25.1.

1. Compute the multiplication law in terms of real and "dual" parts, i.e., $z_1 z_2 = z = x + \varepsilon y$, and find x and y. Compute the addition law for $z_1 + z_2$.
2. Show that $\mathbb{R}(\varepsilon)$ is not a field.
3. Compute the law for all positive powers of $z = x + \varepsilon y$, and for negative powers when $x \neq 0$.
4. Consider $z = x + \varepsilon$ and evaluate its positive powers, and its negative powers when $x \neq 0$. How is each dual part related to the corresponding real part?
5. Let $p(x)$ be a real polynomial and extend it to dual arguments. Show that

$$p(x + \varepsilon) = p(x) + \varepsilon p'(x).$$

6. Let $q(x)$ be a polynomial in negative powers of x. Show, for $x \neq 0$, that $q(x+\varepsilon) = q(x) + \varepsilon q'(x)$.
7. Let $p(x)$ and $q(x)$ be polynomials, and consider the *rational function* $f(x) = p(x)/q(x)$. Where $q(x) \neq 0$, extend the rational function to $\mathbb{R}(\varepsilon)$ and prove that $f(x+\varepsilon) = f(x) + \varepsilon f'(x)$.

25.2 Solutions

25.2.1 Tutor note: Dual numbers

This exercise is perhaps for the specially interested.

Solution to Exercise 25.1

For part 2.

Since $\varepsilon^2 = 0$, ε cannot have an inverse. If $a\varepsilon = 1$, multiplication by ε would give $a\varepsilon^2 = \varepsilon$, hence $0 = \varepsilon$, a contradiction.

For part 3.

Use the binomial formula and $\varepsilon^2 = 0$.

For part 4.

$(x+\varepsilon)^n = x^n + \varepsilon n x^{n-1}$. The dual part is the derivative of the real part.

$$(x+\varepsilon)^{-1} = \frac{x-\varepsilon}{(x-\varepsilon)(x+\varepsilon)} = \frac{x-\varepsilon}{x^2} = x^{-1} - \varepsilon x^{-2}.$$

Computing further,

$$(x+\varepsilon)^{-n} = x^{-n} - \varepsilon n x^{-n-1}.$$

Again, the dual part is the derivative of the real part.

26 Newton's method

Newton's method replaces a nonlinear system by its first-order Taylor approximation at each iteration.

In the next exercise, we derive the Newton–Raphson method for root finding.

Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^n$ be a function of class C^2. We seek $\mathbf{x} \in \Omega$ such that $f(\mathbf{x}) = 0$.

Assume we have an initial guess $\mathbf{x}_0 \in \Omega$.

26.1 Exercises

Exercise 26.1. Explain why the following polynomial is a good approximation to $f(\mathbf{x})$ near $\mathbf{x}_0$:

$$p(\mathbf{x}) = f(\mathbf{x}_0) + J(\mathbf{x} - \mathbf{x}_0), \quad J = Df(\mathbf{x}_0).$$

Exercise 26.2. Newton's method now finds a new guess $\mathbf{x}_1$ by finding the root of the Taylor polynomial, i.e., $p(\mathbf{x}_1) = 0$. Write down a formula for $\mathbf{x}_1$, assuming that the matrix J has an inverse.

Exercise 26.3. Suppose the true solution is $\mathbf{x}_* \in \Omega$, and assume that the error in $\mathbf{x}_0$ is sufficiently small. Find an estimate for the error of $\mathbf{x}_1$. Assuming that the error can be converted to significant digits n, what is the number of significant digits in $\mathbf{x}_1$? Hint: Look at the remainder term.

Exercise 26.4. Suppose successive iterations are performed. How many digits do you have after k iterations?

Exercise 26.5. Make a Python implementation of Newton's method and test it on the following function:

$$f(x, y) = (e^x y^3 - 1, y^2 - \sin(x) - 1).$$

It can be an idea to try and visualize the function. Two roots are:

$$\{(4.38277027, 0.23202192),\ (0, 1)\}$$

Try different initial guesses $\mathbf{x}_0$, close to the roots and further away. How many iterations do you need to achieve machine precision for your guesses? Can you find more roots?

26.2 Solutions

26.2.1 Tutor note: Newton–Raphson method

Tutor note: Newton–Raphson is an important nonlinear root-finding algorithm, although quantum-chemistry calculations may avoid a full Newton step because solving with the Jacobian is expensive. This exercise also develops Taylor approximations of vector-valued functions.

Solution to Exercise 26.1

For a C^2 function, Taylor's theorem gives

$$f(\mathbf{x}) = f(\mathbf{x}_0) + Df(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) + \mathcal{O}(\|\mathbf{x} - \mathbf{x}_0\|^2).$$

Solution to Exercise 26.2

$$\mathbf{x}_1 = \mathbf{x}_0 - J^{-1} f(\mathbf{x}_0).$$

Solution to Exercise 26.3

The source does not supply a solution.

Solution to Exercise 26.4

The source does not supply a solution.

Solution to Exercise 26.5

The source does not supply a solution.

27 Complex-step differentiation

Complex-step differentiation estimates derivatives without the subtractive cancellation of a centered finite difference.

Let $f(x)$ be a real-valued function, assumed to be analytic near some x, i.e., it agrees with its Taylor series in the vicinity of x.

27.1 Exercises

Exercise 27.1.

1. Explain why there exists a complex analytic function near $x + 0\mathrm{i}$ that agrees with f on the real axis.

A classical method for computing a numerical derivative of a C^1 function is the finite difference approximation:

$$f'(x) \approx \delta_h f(x) := \frac{f(x+h) - f(x-h)}{2h}.$$

The error is $\mathcal{O}(h^2)$. In the rest of the exercise, let f be given by

$$f(x) = \exp(x),$$

and we wish to differentiate around $x = 1$.

1. Write a small program that uses standard double precision floating point arithmetic to compute the finite difference derivative for step lengths from the interval $h \in [10^{-9}, 10^{-1}]$.
2. Make a plot of the absolute value of the error in the approximation as function of h. Use log scales.

The *complex step method* utilizes complex analyticity to avoid the cancellation errors seen in the finite difference scheme. It relies on the function in question being implemented/implementable using complex arithmetic.

1. Show that

$$f(x + \mathrm{i}h) = f(x) + \mathrm{i}hf'(x) - \frac{1}{2}h^2 f''(x) - \frac{\mathrm{i}h^3}{6} f'''(x) + \mathcal{O}(h^4).$$

2. Solve for $f'(x)$, and show that

$$f'(x) = \Im \frac{f(x + \mathrm{i}h)}{h} + \mathcal{O}(h^2).$$

   This is the complex step method.

3. In the plot from above, add a plot of the error of the complex step method. Compare.

27.2 Solutions

27.2.1 Tutor note: Complex step method

Tutor note: I think the tutors might be surprised by this exercise! I learned about the method only last year. This exercise is very nice, in my opinion. However, it may be a little advanced for some students, because it requires basic understanding of cancellation errors in floating point arithmetic.

Solution to Exercise 27.1

Source solution 1.

The real power series has a positive radius of convergence. Using the same coefficients for a complex argument gives a complex power series with the same radius of convergence, and hence a unique analytic extension near x.

Source solution 2.

The student should observe that the accuracy deteriorates when h becomes too small, around 10^{-7}.

Source solution 3.

Just compute the Taylor series.

Source solution 4.

The main observation is that the zeroth and second order terms are purely real, while the first and third are purely imaginary. The given approximation then eliminates the real parts, which explains the second-order error.

Source solution 5.

The student should observe that the accuracy does *not* deteriorate as it does for the centered difference. The complex-step formula avoids subtraction of nearly equal function values.

Part X

Fourier analysis

Part XI

Ordinary differential equations

Part XII

Partial differential equations

Part XIII

Optimization